Complexity-based Economic and Production Analysis: Overview and Country Report

With analysis for Brazil, China, Egypt, France, Germany, Japan, Malaysia, Mexico, Morocco, South Africa, South Korea, Spain, Turkey, UAE and USA

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1. Overview

- With an increasing volume of international trade, countries are more specialized than ever in production and have diversified into distinct production patterns
- Based on global export data available to us, in this report we examine countries' growth paths and product spaces on a complexity level, during the period from 2009 to 2014
- The rest of the report is organized as follows:

Section 2	Countries' economic complexity is quantified the same way as defined by Hausmann and Hidalgo (2009)
Section 3	Products are grouped into industry families and the growth paths adopted by countries during the period are closely examined
Section 4	Analyses are conducted on countries' sequential development strategy, sensitivity to industry upheaval and the contingency plan
Section 5	Macro-level results are used to assess countries' structural optimality
Section 7	Country-specific results are available for further reviews

2. Economic Complexity Framework

2.1 Economic Complexity

2.2 Production Matrix

2.3 Diversity and Ubiquity

2.4 Economic & Product Complexity Index

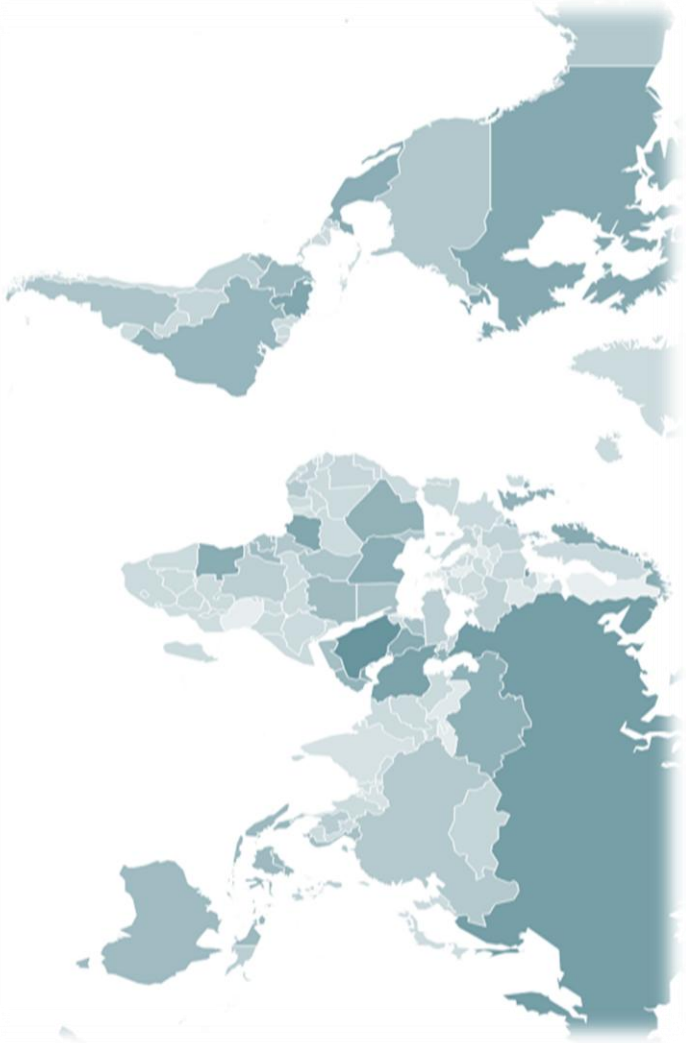
2.5 Proximity and Distance

2.6 Opportunity Gain

2.1 Economic Complexity

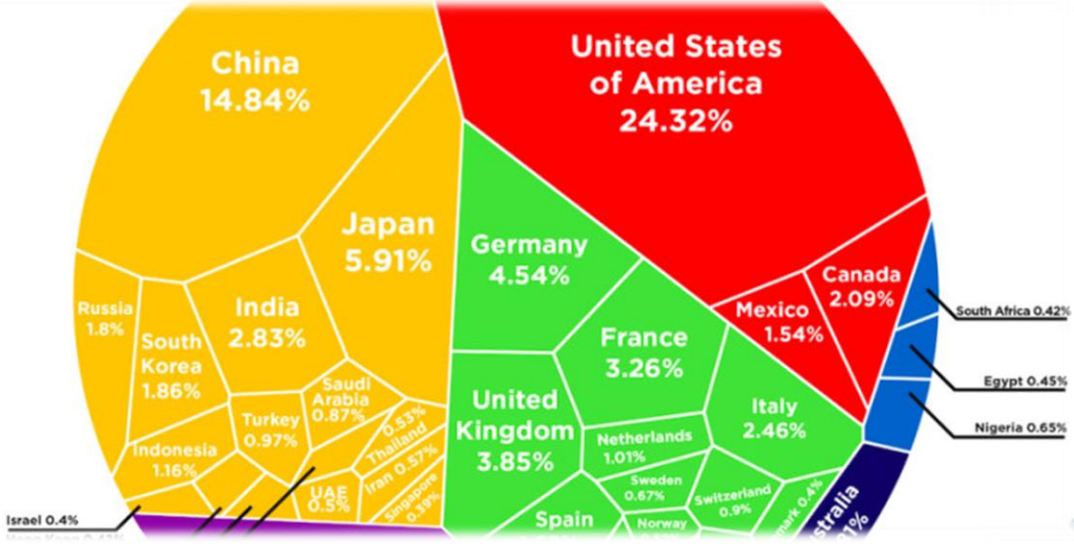
In the era of globalization, more and more economies decide to specialize in the production. The global market and international organizations allow the knowledge that is held by the few to reach many.

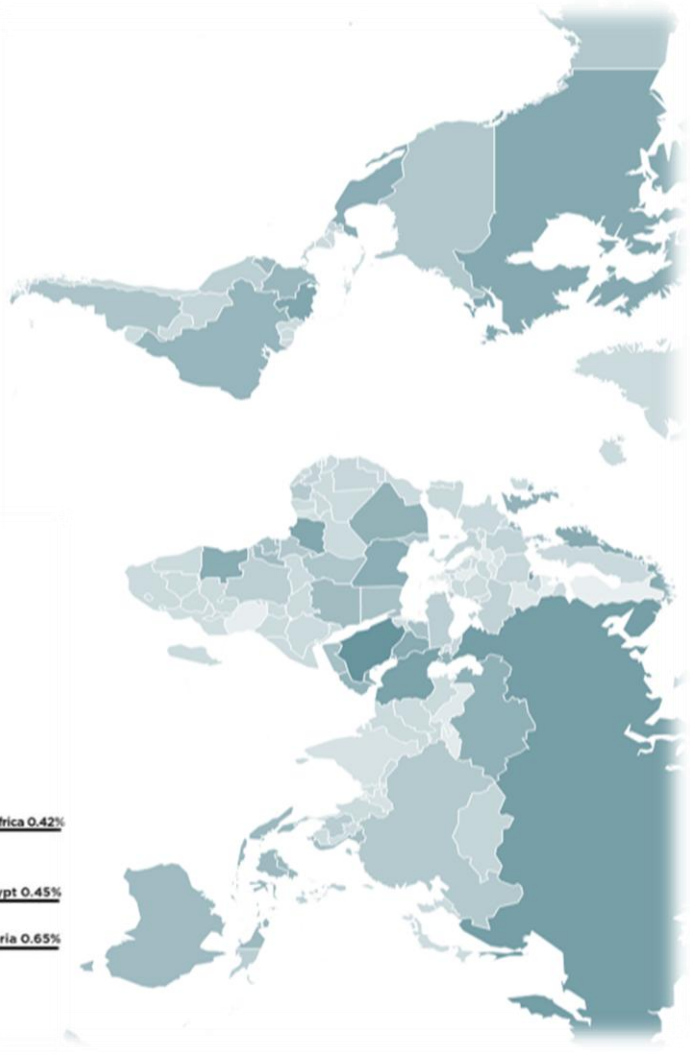




2.1 Economic Complexity

Meanwhile, the uneven distribution of productive knowledge is widening gap between rich and poor economies. The uneven distribution is expressed in the diversity and sophistication of products the economies produce.





2.1 Economic Complexity

There are two types of productive knowledge:

Explicit Knowledge

that can be transferred easily by reading and listening

- Weather forecast
- Bus schedule
- History of a building

Tacit Knowledge

that requires a systematic and time-consuming learning process to transfer

- Semiconductor Manufacturing
- Shell oil drilling
- Foreign Language

A lack of crucial tacit knowledge is the hindrance that an economy faces in its development and growth



2.1 Economic Complexity

Products are made of raw material with the support of capital and machinery. However, from the perspective of Economic Complexity:

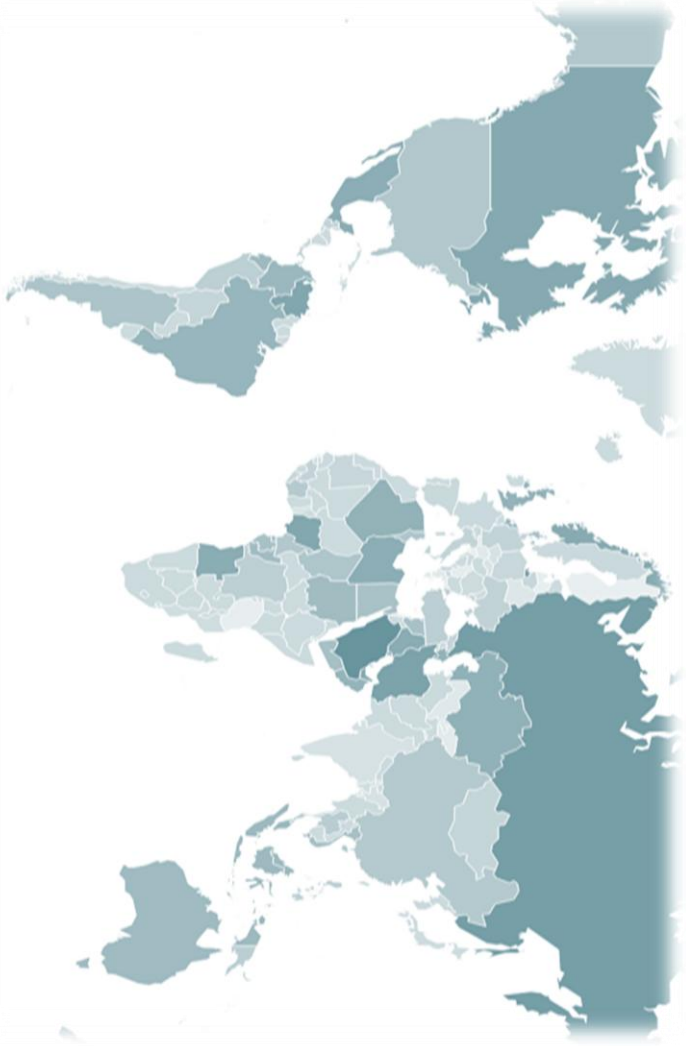
Products are made of pieces of productive knowledge



- Water, pigment substance, and glass
- Manufacturing Machinery



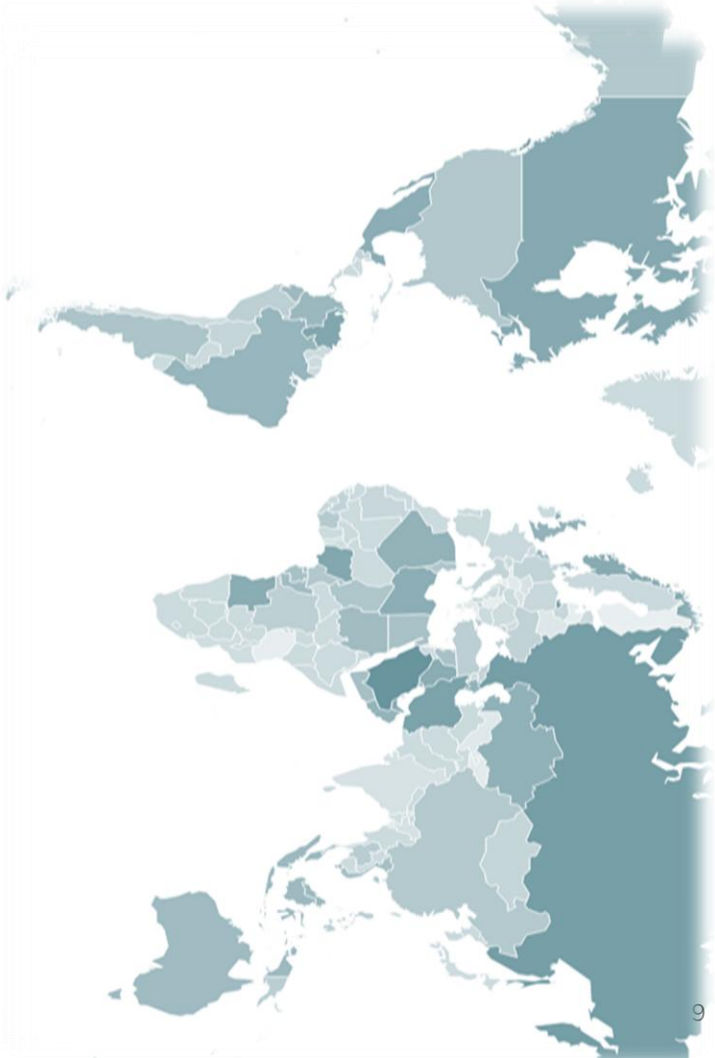
- How to control the tinting strength of pigments
- How to mass produce glass bottles



2.1 Economic Complexity

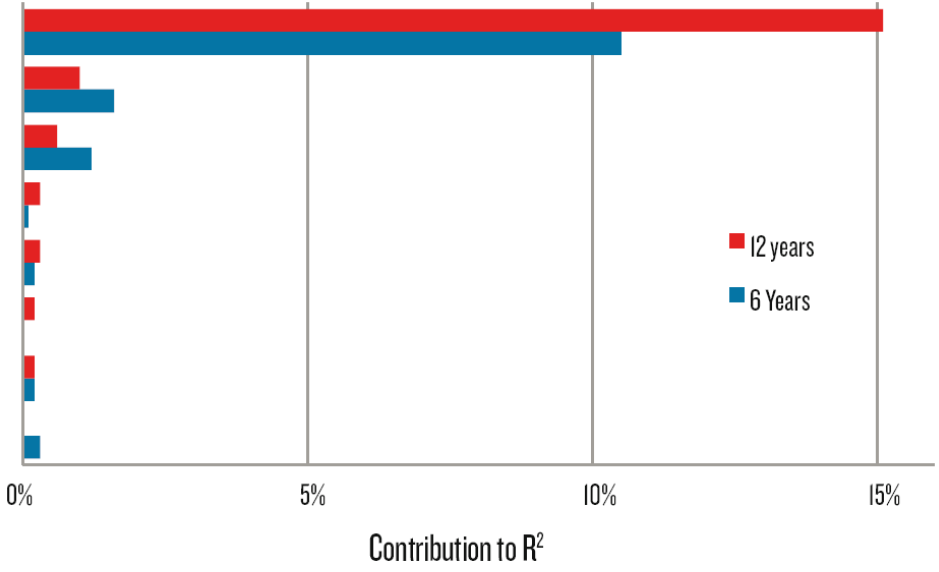
The complexity of an economy is related to the multiplicity of useful knowledge embedded in it.

Economic complexity, therefore, is expressed in the composition of a country’s productive output and reflects the structure that emerges to hold and combine knowledge



2.1 Economic Complexity

The economic complexity more effectively captures growth relevant information than the six world governance indicators.



Economic Complexity Index

All Instituional Variables

Political Stability

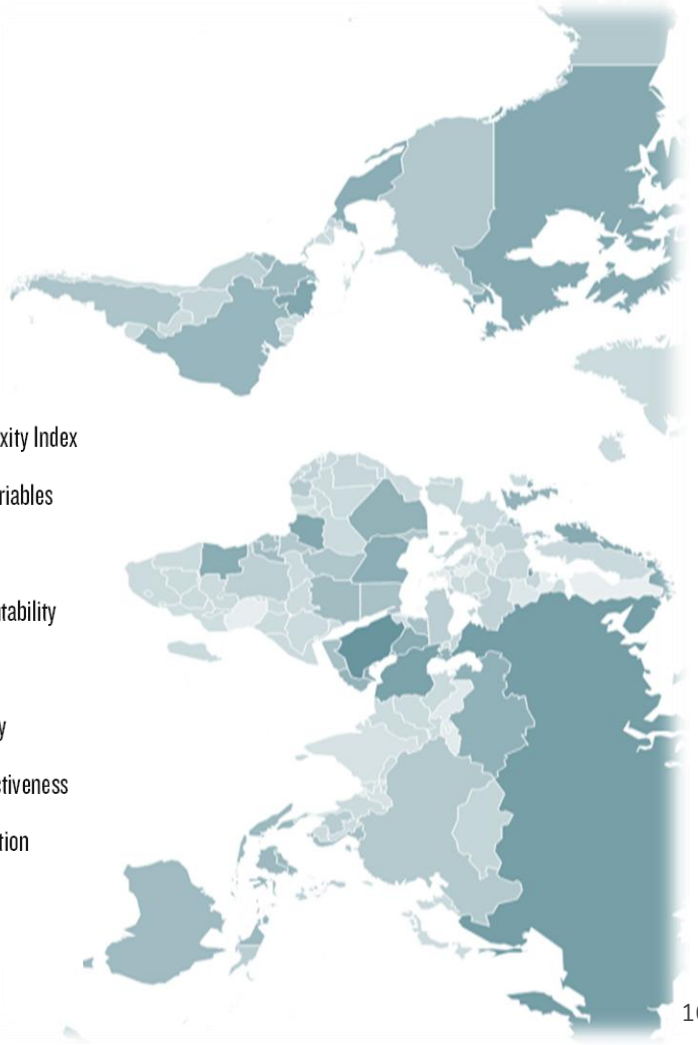
Voice and Accountability

Rule of Law

Regulatory Quality

Government Effectiveness

Control of Corruption



2.2 Production Matrix

- When consider which products play important roles in a country’s production, we should take into account of the export volume of the country and that of the product in the world trade.
- In other words, we should use comparative advantage instead of absolute advantage as our criterion.



2.2 Production Matrix

- Balassa defines Revealed Comparative Advantage (RCA) as *"a country has Revealed Comparative Advantage in a product if it exports more than its "fair" share, that is, a share that is equal to the share of total world trade that the product represents."*

$$RCA_{cp} = \frac{X_{cp}}{\sum_c X_{cp}} / \frac{\sum_p X_{cp}}{\sum_{c,p} X_{cp}}$$

- Accordingly, we define the matrix of comparative advantage in production for countries and products. "1" indicates the advantage, while "0" indicates the opposite

$$M_{cp} = \begin{cases} 1 & RCA_{cp} \geq 1 \\ 0 & Otherwise \end{cases}$$

2.3 Diversity and Ubiquity











2.3 Diversity and Ubiquity

- Diversity measures how many products a country can produce
- Ubiquity measures how many countries are capable to produce a product
- The Diversity and Ubiquity are a pair of complements and can be used to iteratively correct each other

$$Diversity = k_{c,0} = \sum_p M_{cp}$$

$$k_{c,N} = \frac{1}{k_{c,0}} \sum_p M_{cp}\, k_{p,N-1}$$

$$Ubiquity = k_{p,0} = \sum_c M_{cp}$$

$$k_{p,N} = \frac{1}{k_{p,0}} \sum_c M_{cp}\, k_{c,N-1}$$

$$k_{c,N} = \frac{1}{k_{c,0}} \sum_p M_{cp} \frac{1}{k_{p,0}} \sum_{c'} M_{c'p}\, k_{c',N-2}$$

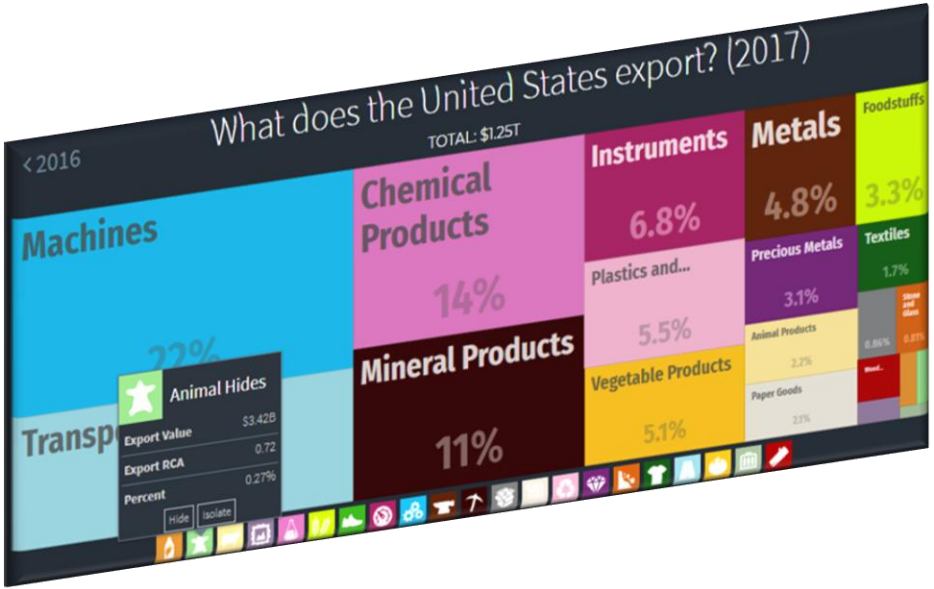
$$k_{c,N} = \sum_{c'} M_{cc'}\, k_{c',N-2}$$

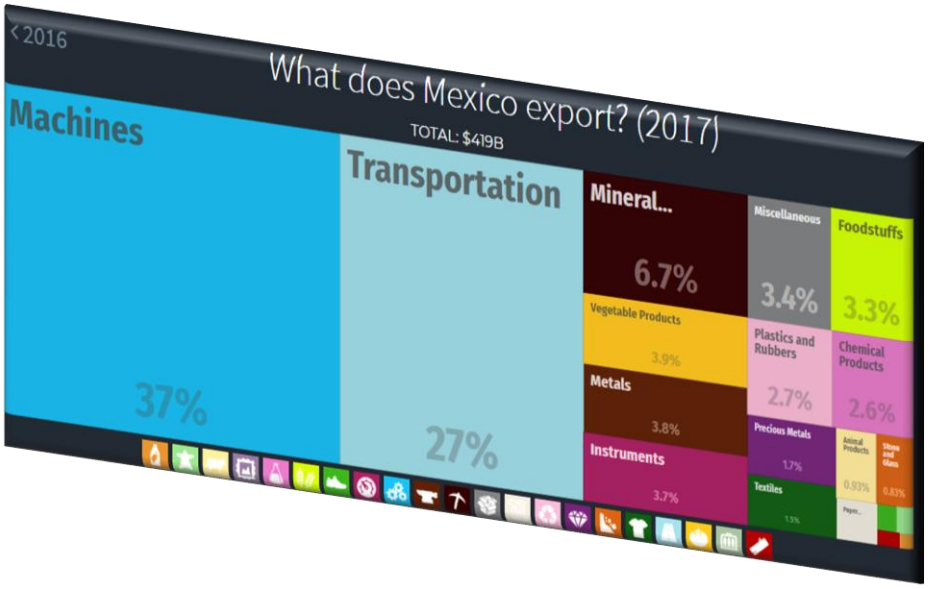
where $M_{cc'} = \sum_p \frac{M_{cp} M_{c'p}}{k_{c,0} k_{p,0}}$

Similarly, we can derive $M_{pp'}$

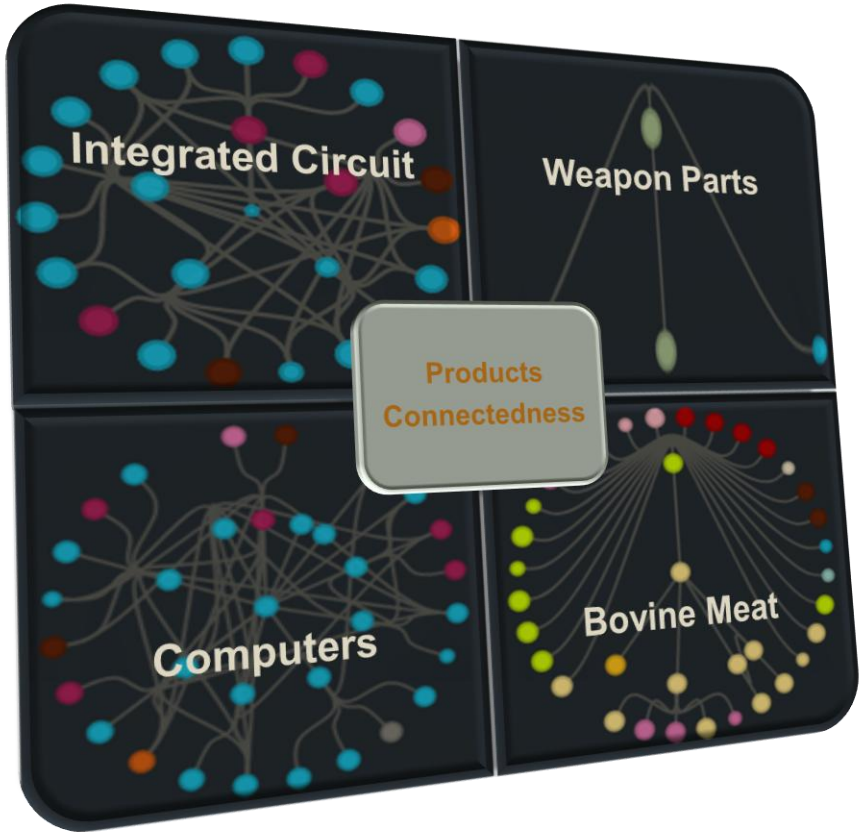
2.4 Economic & Production Complexity Index

Countries specialize in different industries with distinct level of productivity





2.4 Economic & Production Complexity Index

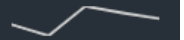


• High complexity products are connected with a large amount of products

• High complexity products are close to other products with high complexity

2.4 Economic & Production Complexity Index

From the trend of Economic Complexity Index in nearby 5 years, we can have a rough picture of whether the countries are on the right track of economic development.

Country	2013	2014	2015	2016	2017	
Japan	2.37352	2.31842	2.29751	2.27406	2.30938	
Switzerland	2.05181	1.99456	2.15805	2.22117	2.24386	
Germany	1.84608	1.81367	2.09809	2.08459	2.07537	
Singapore	1.71717	1.71171	1.746	1.79973	1.86534	
Sweden	1.75214	1.6459	1.92429	1.86277	1.80773	
South Korea	1.82762	1.90646	1.65462	1.69142	1.77613	
United States	1.43702	1.30167	1.8166	1.78168	1.75541	
Finland	1.57477	1.49895	1.77048	1.72464	1.70679	

2.4 Economic & Production Complexity Index

The second largest eigenvector of M_{cc} best preserves the information about variance in the system, which is just Economic Complexity. Hence, we define the Economic Complexity Index (ECI) as:

$$ECI = \frac{\vec{K} - < \vec{K} >}{stdev(\vec{K})}$$

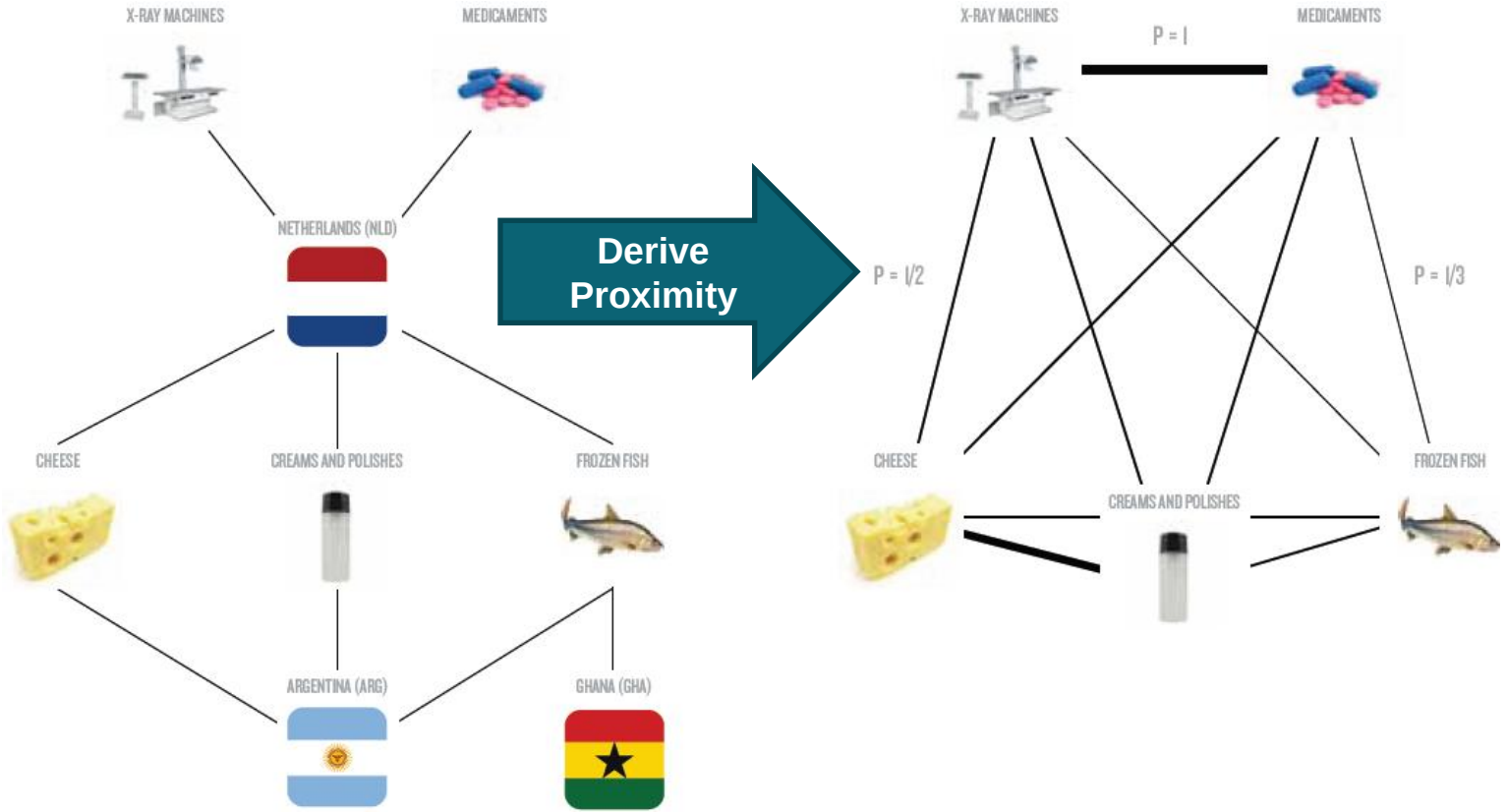
Where <> stands for numeric mean, and stdev stands for the standard deviation.

Similarly, we can calculate the second largest eigenvector of $M_{pp'}$, and define Product Complexity Index (PCI) as

$$PCI = \frac{\vec{Q} - <\vec{Q}>}{stdev\ (\vec{Q})}$$

Where<> stands for numeric mean, and stdev stands for the standard deviation.

2.5 Proximity&Distance



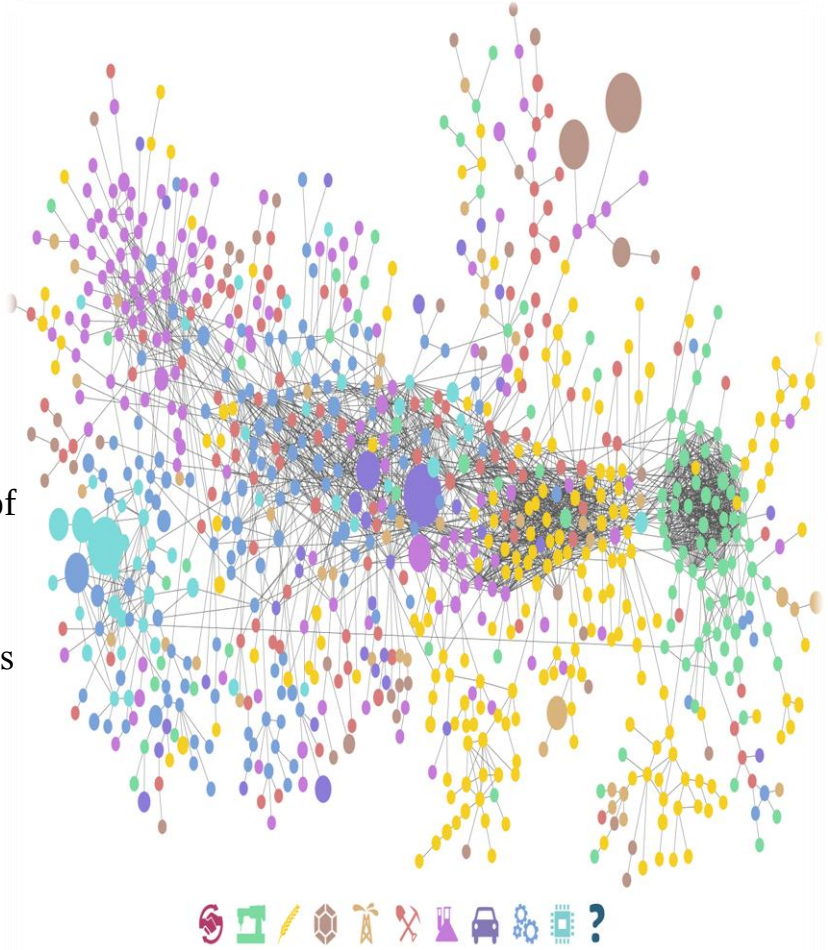
2.5 Proximity&Distance

- Certain products are composed of similar productive knowledge. We define connectedness of z product as the numeric average of proximity of it to all other products

- We define the smaller conditional probability of exporting product p given the country is currently exporting product p’ and that of exporting p’ given the country is exporting p, as proximity. We then normalize Proximity as Distance

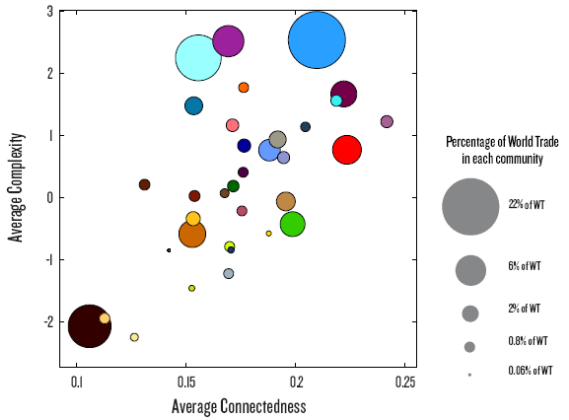
$$\emptyset_{pp'} = \frac{\sum_c M_{cp} M_{cp'}}{max\,(k_{p,0,} k_{p',0})}$$

$$d_{cp} = \frac{\sum_{p'}(1-M_{cp'})\emptyset_{pp'}}{\sum_{p'} \emptyset_{pp'}}$$



2.6 Opportunity Gain

While the product space gives us a glimpse of productive knowledge embedded in an economy, we are still in need of a quantitative measure of the opportunity those productive knowledge implies. We define this measure as opportunity value.



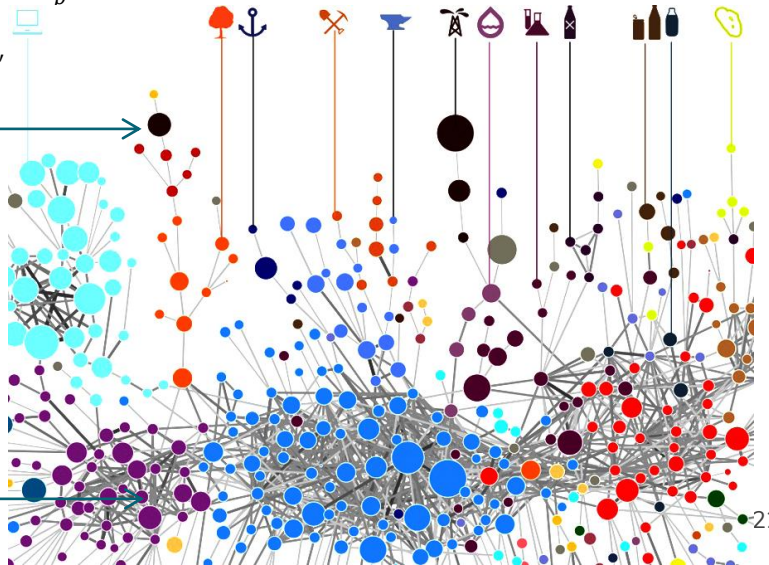
2.6 Opportunity Gain

- The profit of exploring a new product is defined as the decrement of Distance to all other products far away from the product space.The Distances are weighted by the complexity of products respectively

$$Total\ Unexploited\ Value = \sum_{p'}(1 - d_{cp'})(1 - M_{cp'})PCI_{p'}$$

$$Opportunity\ Gain_{cp} = \sum_{p'} \frac{\emptyset_{pp'}}{\sum_{p''} \emptyset_{p''p'}}(1 - M_{cp'})PCI_{p'}$$

However, the low complexity and connectedness of a product can also incur a cost that needs to be deducted from the opportunity gain



3. Production Analysis

3.1 Product Categorization

3.2 Actual Development Index

3.3 Regional and Global Comparison

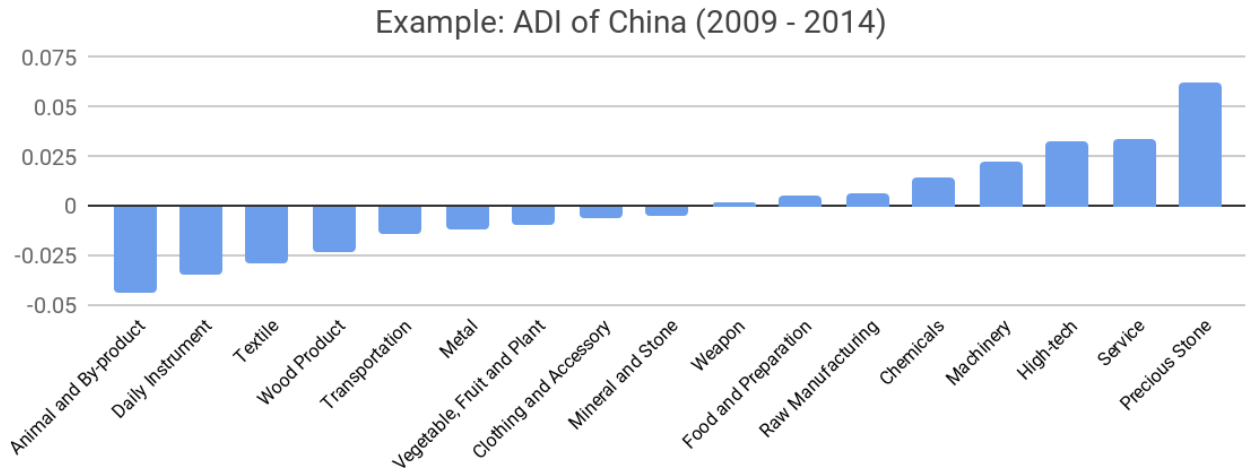
3.1 Product Categorization

- Product groups in UN Comtrade are labelled from 010000-999999, with the first two-digits representing a category to which the product belongs
- From the 99 categories as extracted from the label, we further group the products into 18 major and widely-accepted industry categories
- High-tech is an aggregated industry that indicates the most complicated products from the most technologically-advanced industries, as measured by PCI, and in regression it explains over 67.6% of countries' GDP

Animal and By-product	Raw Manufacturing	Transportation	Vegetable, Fruit and Plant
Wood Product	Weapon	Food and Preparation	Clothing and Accessory
Daily Instrument	Mineral and Stone	Textile	Precious Stone
Chemical	Metal	Service	Chemical Manufacturing
Machinery	High-tech		

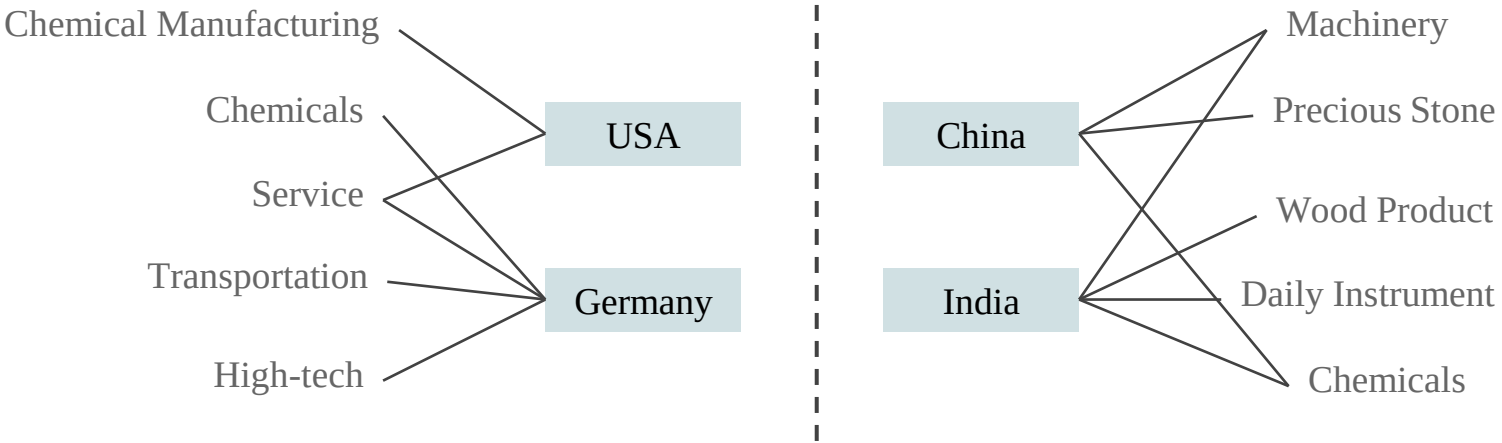
3.2 Actual Development Index

- Based on this concept of distance between countries and product families, we defining the net distance change between two periods as the Actual Developmental Index (ADI)
- ADI measures by how much a country has moved in the direction of a certain product family in its product space during the period, and is a relative concept



3.3 Regional and Global Comparison

- It can be observed that leading economies, such as U.S. and Germany, in general focus on products with higher level of complexity
- Reasonably, emerging economies such as China, South Korea, Singapore and India have devoted the majority of their efforts into intermediate industry groups



4. Production Space Design

4.1 Suggested Development Vector

4.2 Sequential Development Strategy

4.3 Sensitivity Analysis

4.4 Contingency Plan

4.1 Suggested Development Vector

- We define the ranking of opportunity gain of a country as the Suggested Development Vector (SDV) of the country
- Leading economies are typically well-shaped and barely have any industries with positive opportunity gain, while emerging economies need a sequential strategy that allows for prioritization of certain industries

Ranking	Brazil	Mexico	China	India
1	Precious Stone	Precious Stone	Machinery	High-tech
2	Service	Daily Instrument	High-tech	Machinery
3	Clothing and Accessory	High-tech	Chemicals	Daily Instrument
4	Daily Instrument	Machinery	Daily Instrument	Transportation
5	High-tech	Mineral and Stone	Textile	Chemical Manufacturing
6	Chemical Manufacturing	Vegetable, Fruit and Plant	Metal	Weapon
7	Machinery	Food and Preparation	Raw Manufacturing	Wood Product
8	Textile	Transportation	Wood Product	Metal
9	Chemicals	Animal and By-product	Clothing and Accessory	Raw Manufacturing
10	Metal	Clothing and Accessory	Service	Animal and By-product

4.2 Sequential Development Strategy

- As the difficulty of bumping up MCP varies across industries, we need to capture this variation of difficulty by defining the concept of marginal opportunity gains as follows

$$Marginal\ Opportunity\ Gain_{cp} = \frac{Opportunity\ Gain_{cp}}{\left(\frac{\sum_p X_{cp} * \sum_c X_{cp}}{\sum_{c,p} X_{cp}} - X_{cp}\right)} = \frac{Opportunity\ Gain_{cp}}{\left(\frac{\sum_p X_{cp} * \sum_c X_{cp}}{X_{cp} * \sum_{c,p} X_{cp}} - 1\right)}$$

- In each iteration, we order all 18 industries based on their current marginal opportunity gain, and prioritize the industry with the highest marginal opportunity gain
- We then increase the prioritized industry's export to an extent that the MCP firmly bumps up to 1, and re-perform calculations to prioritize the next industry

4.2 Sequential Development Strategy

- Calculated using the sequential strategy algorithm, the example of Morocco shows how the prioritization of a developing country evolves from basic products to intermediate-level products

Before Iteration	Iteration 1	Iteration 2	Iteration 3
Weapon	Textile	Wood Product	Food and Preparation
Textile	Daily Instrument	Chemical Manufacturing	Chemical Manufacturing
Daily Instrument	Wood Product	Daily Instrument	Precious Stone

Iteration 4	Iteration 5	Iteration 6	Iteration 7
Precious Stone	Chemical Manufacturing	Daily Instrument	Raw Manufacturing
Chemical Manufacturing	Daily Instrument	Raw Manufacturing	Metal
Daily Instrument	Raw Manufacturing	Metal	Mineral and Stone

Iteration 8	Iteration 9	Iteration 10	Iteration 11
Metal	Mineral and Stone	Transportation	Animal and By-product
Mineral and Stone	Transportation	Animal and By-product	Vegetable, Fruit and Plant
Transportation	Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation

4.3 Sensitivity Analysis

- The sensitivity analysis aims to approximate the volatility of SDV subject to an increment or decrement of export and is mainly implemented on developed countries that have few industries with positive opportunity gains.
- To be more specific, sensitivity score is the least-squared slope for the linear regression between percentage change of production and that of the E_f CI
- We perform the sensitivity analysis of the aggregate of top ECI countries, including US, UK, Korea, Japan, South Korea, Singapore, Germany, and France, and use the result as the benchmark. The sensitivity score in 2014 and 2009 are shown below:

Sensitivity Score (2014)		
	High-tech	Food and Preparation
Decremental	1.676	0.007
Incremental	1.195	-0.006

Sensitivity Score of Benchmark in 2014

Sensitivity Score (2009)		
	High-tech	Food and Preparation
Decremental	1.514	-0.001
Incremental	1.294	0.007

Sensitivity Score of Benchmark in 2009

4.4 Contingency Plan

The Contingency Plan presents solutions to productivity decline within specific industries. Sometimes, the optimal solution to such downturns is not to invest in the industries undergoing downturns directly, but to support industries highly connected to such industries.

Contingency Plan (2014)		
	0	1
High-tech	Animal and By-product	Precious Stone
Service	Service	Precious Stone

Contingency Plan of Benchmark in 2014

Contingency Plan (2009)		
	0	1
High-tech	Mineral and Stone	Precious Stone
Service	Mineral and Stone	Precious Stone

Contingency of Benchmark in 2009

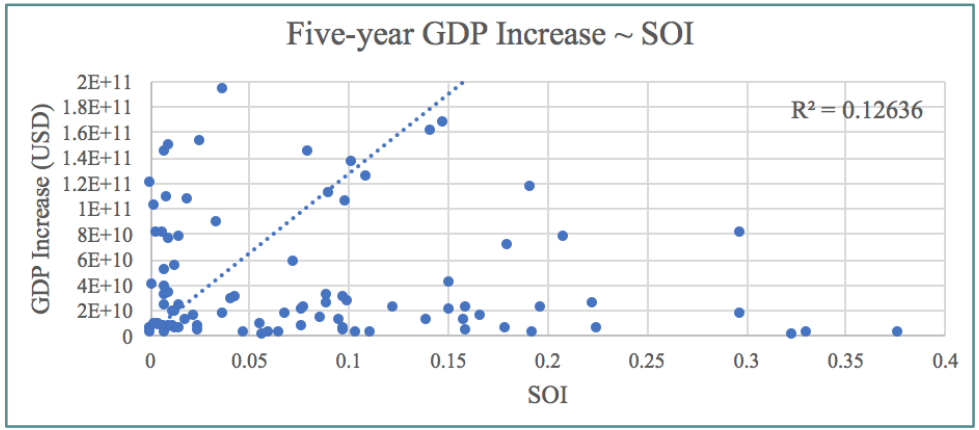
5. Macro-level Predictions

5.1 Structural Optimality Index (SOI)

5.2 SOI Examples and Implications

5.1 Structural Optimality Index (SOI)

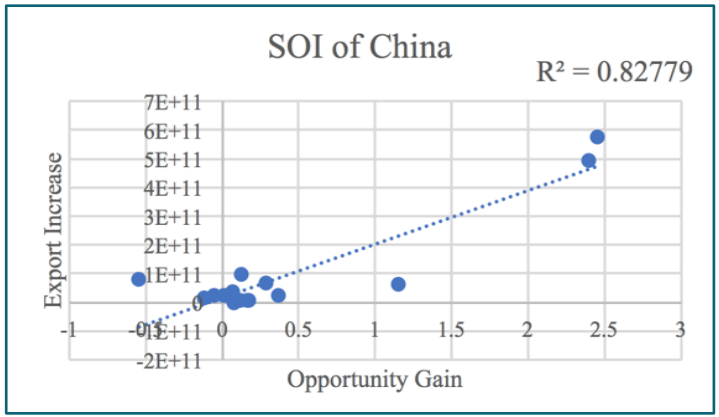
- Structural Optimality Index (SOI) is defined as the adjusted R-Square of export increase in industries ~ opportunity gains of industries
- SOI provides a quantified measure of how the development path adopted by a country fitted with/differed from calculations on a complexity level (i.e., the opportunity gains)
- We further observe a significant correlation between SOI and five-year GDP increase of countries, indicating that structural optimality does impact economic outcomes

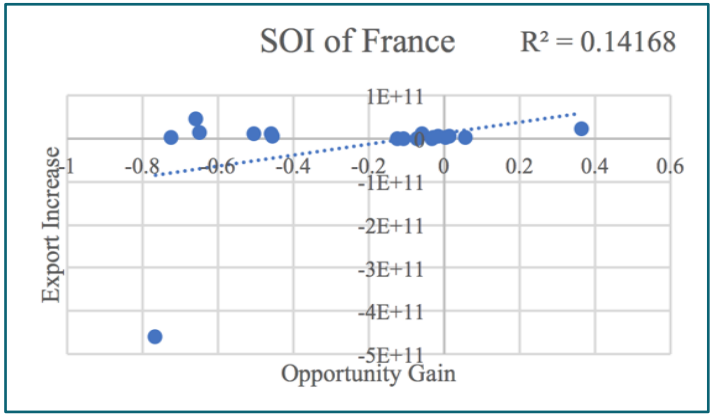


Statistics	Value
Multiple R	0.355
R-Square	0.126
Adjusted R^2	0.119
Variable	**Value**
Coefficient	$1.249*10^{12}$
t-stat	4.061
P-value	0.000(***)

5.2 SOI Examples and Implications

- Taking China and France for comparison, China ranks the 1st place among all countries in terms of SOI (82.8%), and has the highest economic growth rate from 2009 to 2014
- In contrast, developed economies such as France, where there is a moderate SOI of around 14.2% and a corresponding annual growth rate of 1.1%
- Intermediate-term economic progress can be at least partially explained by the optimality of development strategy that countries choose to adopt





6. Reference

- Hausmann, R., Hidalgo, C., Bustos, S., Coscia, M., Chung, S., Jimenez, J., Simoes, A. and Yıldırım, M. (2007). *The Atlas of Economic Complexity*. [online] Atlas.cid.harvard.edu. Available at: http://atlas.cid.harvard.edu/.

- Comtrade.un.org. (n.d.). *Download trade data | UN Comtrade: International Trade Statistics*. [online] Available at: https://comtrade.un.org/data/.

7. Appendix

- In this section we outline the most important calculations & analysis results for a selected group of countries, including Brazil, China, Egypt, France, Germany, Japan, Malaysia, Mexico, Morocco, South Africa, South Korea, Spain, Turkey, UAE and USA

- For each country, we highlight its GDP & ECI trend, Actual Development Index (ADI), Suggested Development Vector (SDV), sensitivity score & plan, contingency table and Structural Optimality Index (SOI) during the period

7.1 China

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Precious Stone	0.062	2. Service	0.034
3. High-tech	**0.033**	**4. Machinery**	**0.023**
5. Chemicals	0.015	6. Raw Manufacturing	0.006
7. Food and Preparation	**0.005**	**8. Weapon**	**0.002**
9. Mineral and Stone	-0.005	10. Clothing and Accessory	-0.006
11. Vegetable, Fruit and Plant	**-0.009**	**12. Metal**	**-0.012**
13. Transportation	-0.014	14. Wood Product	-0.023
15. Textile	**-0.029**	**16. Daily Instrument**	**-0.035**
17. Animal and By-product	-0.044	18. Chemical Manufacturing	-0.063

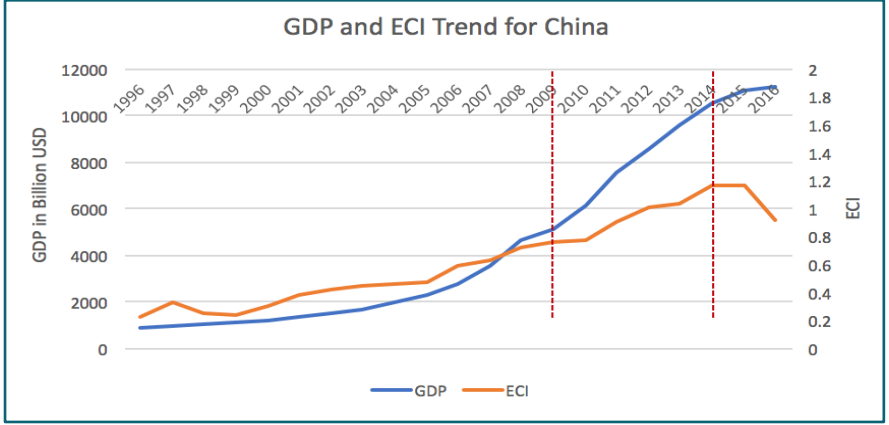


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Transportation	-0.363
Food and Preparation	-0.173
Vegetable, Fruit and Plant	-0.162
Mineral and Stone	-0.119
Animal and By-product	-0.078
Weapon	-0.073
Precious Stone	-0.066
Chemical Manufacturing	-0.059
Service	-0.009
Chemicals	0
Raw Manufacturing	0
Wood Product	0
Clothing and Accessory	0
Textile	0
Metal	0
Machinery	0
Daily Instrument	0
High-tech	0

7.1 China

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Clothing and Accessory	Service	Service	Service	Service	Service
2	High-tech	Metal	Metal	Metal	Metal	Animal and By-product
3	Chemicals	Chemicals	Chemicals	Chemicals	Chemicals	Chemical Manufacturing
4	Daily Instrument	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Precious Stone
5	Raw Manufacturing	Wood Product	Wood Product	Wood Product	Wood Product	Daily Instrument
6	Wood Product	High-tech	High-tech	High-tech	High-tech	Weapon
7	Machinery	Textile	Textile	Textile	Textile	Machinery
8	Metal	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Metal
9	Textile	Machinery	Machinery	Machinery	Machinery	Textile
10	Service	Weapon	Weapon	Weapon	Weapon	High-tech
11	Chemical Manufacturing	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Wood Product
12	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Raw Manufacturing
13	Weapon	Chemical Manufacturing	Chemical Manufacturing	Animal and By-product	Chemical Manufacturing	Chemicals
14	Animal and By-product	Animal and By-product	Animal and By-product	Chemical Manufacturing	Animal and By-product	Clothing and Accessory
15	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation
18	Transportation	Transportation	Transportation	Transportation	Transportation	Transportation
EffCI	1.685	1.538	1.542	1.543	1.455	1.243

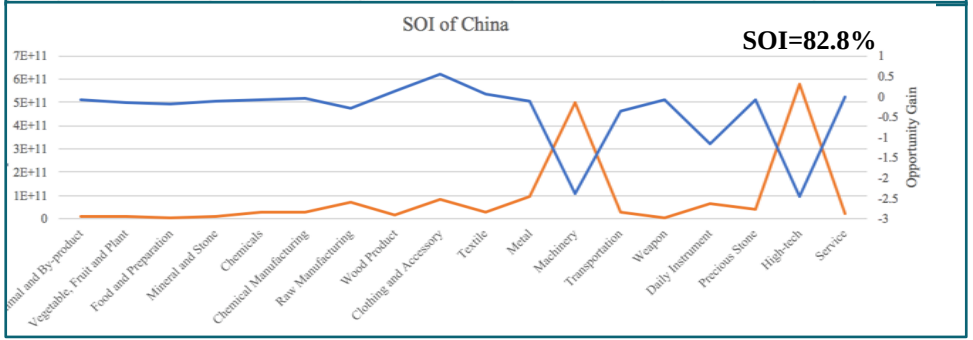


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	1.200	-0.029
Incremental	0.395	-0.035

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	1.265	0.000
Incremental	0.488	-0.077

Table 6: Contingency Plan (2009)

	0	1
Machinery	Service	Animal and By-product
High Tech	Service	Animal and By-product

Table 7: Contingency Plan (2014)

	0	1
Machinery	Mineral and Stone	Weapon
High Tech	Mineral and Stone	Weapon
Daily Instrument	Mineral and Stone	Precious Stone

7.2 France

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Raw Manufacturing	-0.080	2. Mineral and Stone	-0.082
3. Precious Stone	**-0.086**	**4. Chemical Manufacturing**	**-0.095**
5. Chemicals	-0.102	6. Wood Product	-0.111
7. Clothing and Accessory	**-0.113**	**8. Service**	**-0.114**
9. Weapon	-0.114	10. Animal and By-product	-0.118
11. Vegetable, Fruit and Plant	**-0.120**	**12. High-tech**	**-0.129**
13. Food and Preparation	-0.134	14. Metal	-0.136
15. Textile	**-0.147**	**16. Transportation**	**-0.178**
17. Daily Instrument	-0.241	18. Machinery	-0.324

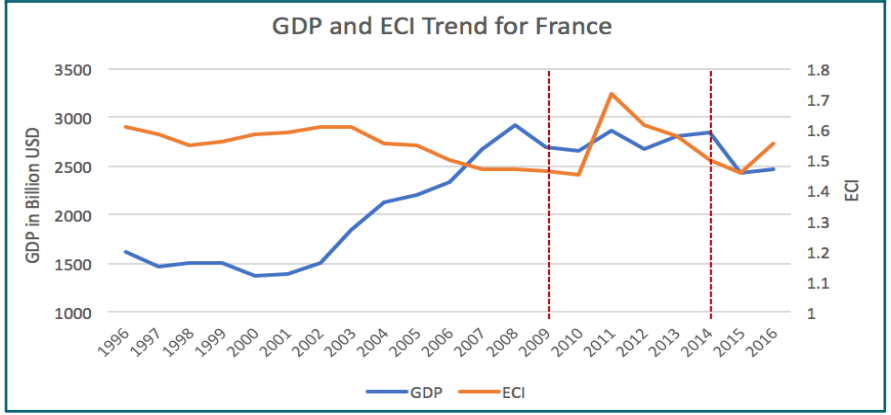


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Mineral and Stone	-0.055
Precious Stone	-0.012
Vegetable, Fruit and Plant	-0.004
Animal and By-product	0
Food and Preparation	0
Chemical Manufacturing	0
Transportation	0
Service	0
Metal	0.015
Clothing and Accessory	0.030
Wood Product	0.031
Raw Manufacturing	0.057
Weapon	0.072
Textile	0.108
Chemicals	0.123
Daily Instrument	0.455
Machinery	0.649
High-tech	0.660

7.2 France

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Service	High-tech	High-tech	High-tech	High-tech	High-tech
2	High-tech	Machinery	Machinery	Machinery	Machinery	Machinery
3	Food and Preparation	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument
4	Chemical Manufacturing	Textile	Textile	Textile	Textile	Textile
5	Raw Manufacturing	Weapon	Weapon	Weapon	Weapon	Weapon
6	Animal and By-product	Service	Service	Service	Chemicals	Chemicals
7	Transportation	Food and Preparation	Food and Preparation	Food and Preparation	Service	Service
8	Mineral and Stone	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing	Raw Manufacturing
9	Clothing and Accessory	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Food and Preparation	Food and Preparation
10	Chemicals	Animal and By-product	Animal and By-product	Animal and By-product	Chemical Manufacturing	Chemical Manufacturing
11	Weapon	Transportation	Transportation	Transportation	Animal and By-product	Animal and By-product
12	Textile	Chemicals	Chemicals	Clothing and Accessory	Transportation	Transportation
13	Vegetable, Fruit and Plant	Clothing and Accessory	Clothing and Accessory	Chemicals	Clothing and Accessory	Clothing and Accessory
14	Precious Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
15	Wood Product	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
16	Metal	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
17	Daily Instrument	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Precious Stone
18	Machinery	Metal	Metal	Metal	Metal	Metal
EtCI	1.224	0.606	0.606	0.614	0.601	0.603

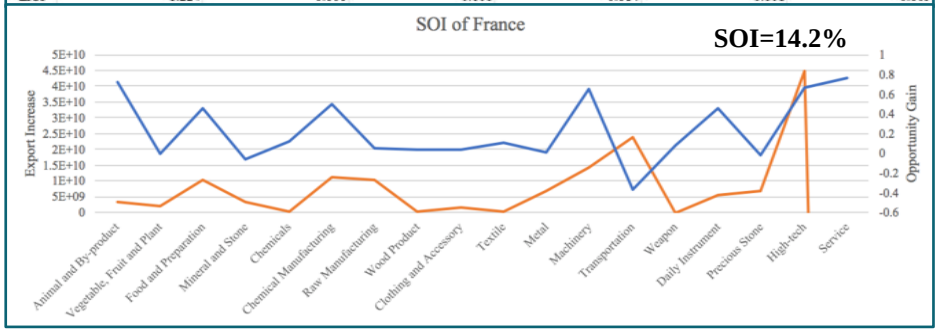


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	-0.147	-0.231
Incremental	0.526	0.067

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	0.469	-0.417
Incremental	0.329	-0.285

Table 6: Contingency Plan (2009)

	0	1
Precious Stone	High-tech	Machinery
High Tech	Machinery	High-tech
Service	Service	Mineral and Stone

Table 7: Contingency Plan (2014)

	0	1
High Tech	Mineral and Stone	Precious Stone
Service	Mineral and Stone	Precious Stone

7.3 Germany

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Chemicals	-0.017	2. Service	-0.025
3. High-tech	**-0.031**	**4. Transportation**	**-0.033**
5. Food and Preparation	-0.040	6. Machinery	-0.041
7. Vegetable, Fruit and Plant	**-0.048**	**8. Raw Manufacturing**	**-0.051**
9. Weapon	-0.052	10. Wood Product	-0.056
11. Precious Stone	**-0.066**	**12. Mineral and Stone**	**-0.072**
13. Clothing and Accessory	-0.073	14. Textile	-0.081
15. Animal and By-product	**-0.086**	**16. Metal**	**-0.097**
17. Chemical Manufacturing	-0.102	18. Daily Instrument	-0.116

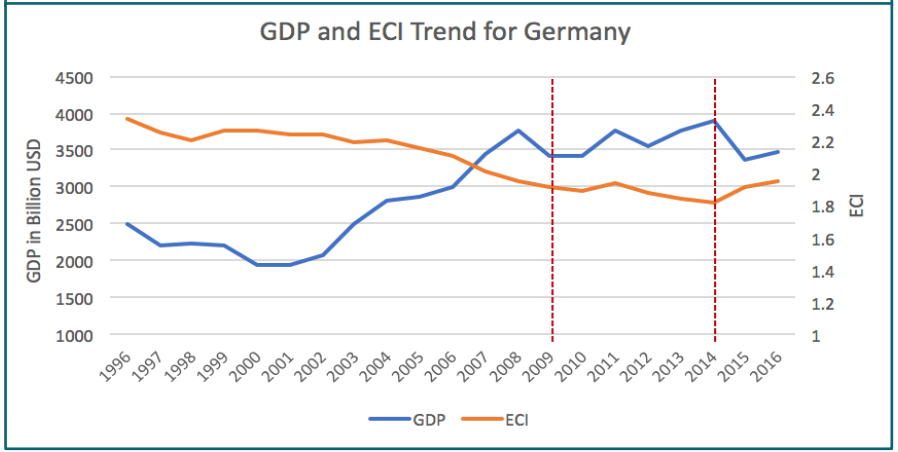


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Vegetable, Fruit and Plant	-0.299
Food and Preparation	-0.291
Textile	-0.234
Precious Stone	-0.230
Animal and By-product	-0.220
Mineral and Stone	-0.205
Clothing and Accessory	-0.200
Chemicals	-0.183
Service	-0.153
Weapon	-0.144
Chemical Manufacturing	0
Raw Manufacturing	0
Wood Product	0
Metal	0
Machinery	0
Transportation	0
Daily Instrument	0
High-tech	0

7.3 Germany

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
2	Chemicals	Chemicals	Chemicals	Animal and By-product	Animal and By-product	Precious Stone
3	Daily Instrument	Daily Instrument	Daily Instrument	Chemicals	Chemicals	Weapon
4	Transportation	Transportation	Transportation	Daily Instrument	Daily Instrument	Animal and By-product
5	Machinery	Machinery	Machinery	Transportation	Transportation	Chemicals
6	Metal	Metal	Metal	Machinery	Machinery	Daily Instrument
7	High-tech	High-tech	High-tech	Metal	Metal	Transportation
8	Wood Product	Wood Product	Wood Product	High-tech	High-tech	Machinery
9	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Wood Product	Wood Product	Metal
10	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing	Raw Manufacturing	Food and Preparation
11	Precious Stone	Precious Stone	Precious Stone	Chemical Manufacturing	Chemical Manufacturing	Wood Product
12	Weapon	Weapon	Weapon	Precious Stone	Precious Stone	Raw Manufacturing
13	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Weapon	Weapon	Chemical Manufacturing
14	Service	Service	Service	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
15	Animal and By-product	Animal and By-product	Animal and By-product	Service	Service	Service
16	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Textile	Textile	Textile	Food and Preparation	Food and Preparation	Textile
18	Food and Preparation	Food and Preparation	Food and Preparation	Textile	Textile	High-tech
ERCI	1.910	1.925	1.933	1.688	1.673	1.239

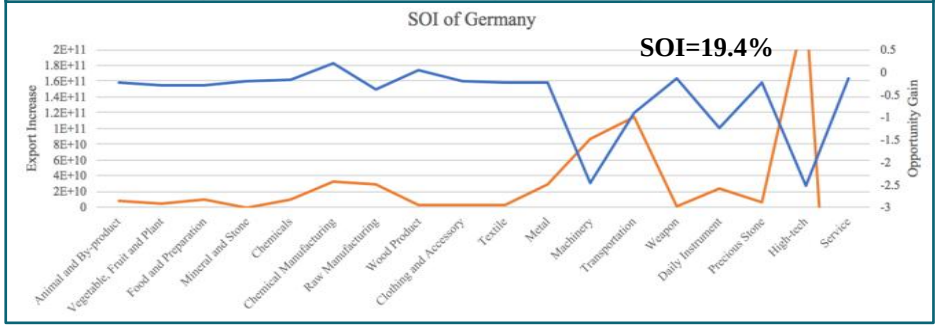


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	2.190	-0.010
Incremental	0.771	-0.135

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	1.341	0.022
Incremental	0.865	-0.054

Table 6: Contingency Plan (2009)

	0	1
High Tech	Clothing and Accessory	Precious Stone
Service	Clothing and Accessory	Service

Table 7: Contingency Plan (2014)

	0	1
High Tech	Mineral and Stone	Precious Stone

7.4 Japan

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. High-tech	0.115	2. Machinery	0.103
3. Transportation	**0.090**	**4. Mineral and Stone**	**0.087**
5. Weapon	0.087	6. Food and Preparation	0.074
7. Wood Product	**0.067**	**8. Service**	**0.067**
9. Vegetable, Fruit and Plant	0.065	10. Metal	0.062
11. Chemicals	**0.060**	**12. Clothing and Accessory**	**0.059**
13. Textile	0.044	14. Precious Stone	0.043
15. Animal and By-product	**0.040**	**16. Chemical Manufacturing**	**0.016**
17. Daily Instrument	0.015	18. Raw Manufacturing	-0.001

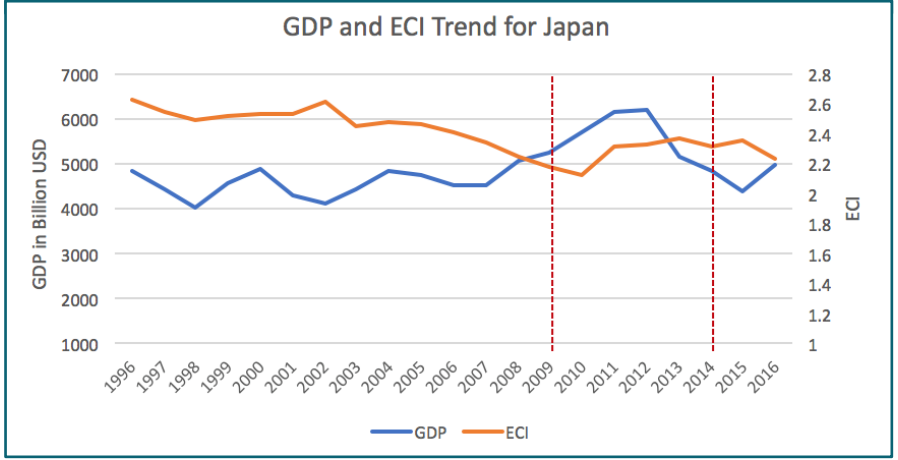


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Food and Preparation	-0.395
Vegetable, Fruit and Plant	-0.392
Wood Product	-0.376
Animal and By-product	-0.359
Precious Stone	-0.316
Clothing and Accessory	-0.303
Textile	-0.302
Service	-0.294
Mineral and Stone	-0.251
Chemical Manufacturing	-0.199
Weapon	-0.197
Chemicals	0
Raw Manufacturing	0
Metal	0
Machinery	0
Transportation	0
Daily Instrument	0
High-tech	0

7.4 Japan

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Raw Manufacturing	Service
2	Transportation	Transportation	Transportation	Transportation	Daily Instrument	Raw Manufacturing
3	Machinery	Machinery	Machinery	Machinery	Transportation	Daily Instrument
4	Metal	Metal	Metal	Metal	Machinery	Transportation
5	High-tech	High-tech	High-tech	High-tech	Metal	Machinery
6	Chemicals	Chemicals	Chemicals	Chemicals	High-tech	Metal
7	Weapon	Weapon	Weapon	Weapon	Chemicals	Chemicals
8	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
9	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Clothing and Accessory	Weapon	Weapon
10	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Chemical Manufacturing	Clothing and Accessory	Chemical Manufacturing
11	Precious Stone	Precious Stone	Textile	Textile	Precious Stone	Textile
12	Textile	Textile	Precious Stone	Precious Stone	Service	Clothing and Accessory
13	Service	Service	Service	Service	Chemical Manufacturing	Precious Stone
14	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Textile	Animal and By-product
15	Wood Product	Wood Product	Wood Product	Wood Product	Animal and By-product	Wood Product
16	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Wood Product	Vegetable, Fruit and Plant
17	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Vegetable, Fruit and Plant	Food and Preparation
18	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation	High-tech
ECI	2.491	2.491	2.516	2.506	2.282	1.521

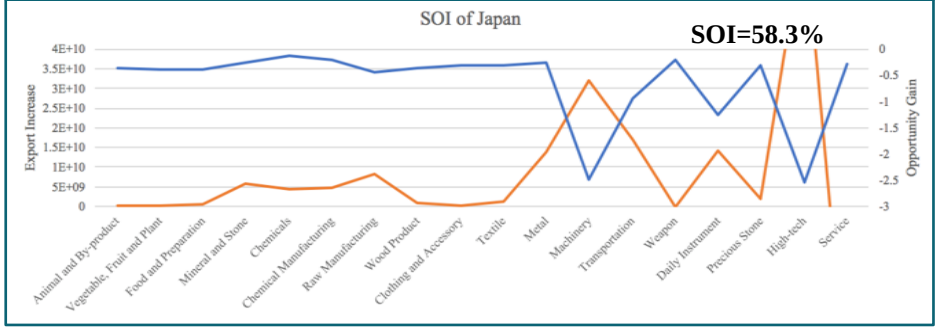


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	1.483	0.000
Incremental	0.632	0.000

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	1.864	0.000
Incremental	0.763	-0.003

7.5 Malaysia

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Precious Stone	0.015	2. Mineral and Stone	0.010
3. Vegetable, Fruit and Plant	**-0.018**	**4. Chemicals**	**-0.023**
5. Food and Preparation	-0.024	6. Weapon	-0.028
7. Animal and By-product	**-0.031**	**8. Metal**	**-0.032**
9. Daily Instrument	-0.033	10. Clothing and Accessory	-0.043
11. Wood Product	**-0.044**	**12. Raw Manufacturing**	**-0.048**
13. Transportation	-0.066	14. Textile	-0.077
15. Chemical Manufacturing	**-0.088**	**16. Machinery**	**-0.099**
17. Service	-0.107	18. High-tech	-0.114

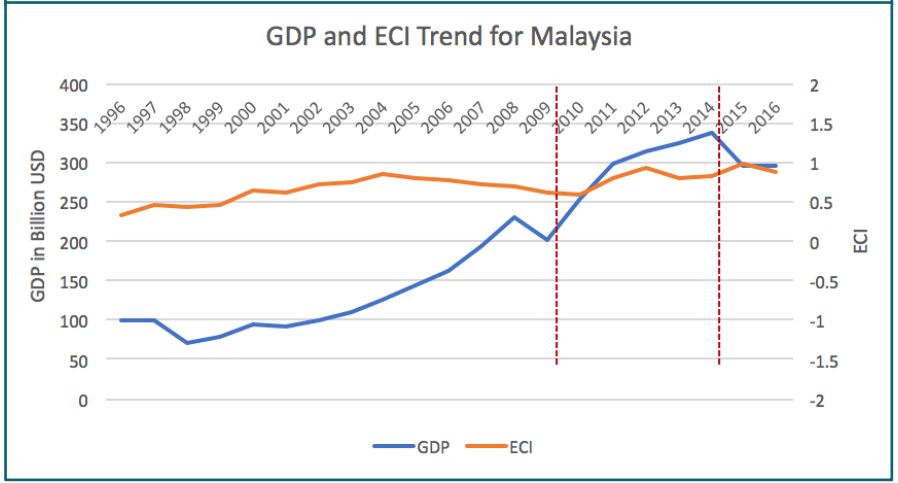


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Service	-0.247
High-tech	-0.246
Machinery	-0.228
Mineral and Stone	-0.226
Wood Product	-0.220
Weapon	-0.209
Transportation	-0.190
Food and Preparation	-0.189
Chemical Manufacturing	-0.187
Raw Manufacturing	-0.146
Clothing and Accessory	-0.117
Precious Stone	-0.108
Metal	-0.107
Vegetable, Fruit and Plant	0
Animal and By-product	0
Textile	0
Chemicals	0
Daily Instrument	0

7.5 Malaysia

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Precious Stone	Precious Stone
2	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Raw Manufacturing	Weapon
3	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Daily Instrument	Animal and By-product
4	Machinery	Machinery	Machinery	Machinery	Vegetable, Fruit and Plant	Service
5	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Machinery	Wood Product
6	High-tech	High-tech	High-tech	High-tech	Metal	Clothing and Accessory
7	Chemicals	Chemicals	Chemicals	Chemicals	Mineral and Stone	Textile
8	Precious Stone	Precious Stone	Precious Stone	Precious Stone	High-tech	Daily Instrument
9	Weapon	Weapon	Weapon	Weapon	Chemicals	Machinery
10	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Weapon	Metal
11	Service	Service	Service	Service	Animal and By-product	Raw Manufacturing
12	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Chemicals
13	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Service	Mineral and Stone
14	Metal	Metal	Metal	Metal	Food and Preparation	Food and Preparation
15	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Wood Product	Vegetable, Fruit and Plant
16	Textile	Textile	Textile	Textile	Chemical Manufacturing	Chemical Manufacturing
17	Wood Product	Wood Product	Wood Product	Wood Product	Textile	Transportation
18	Transportation	Transportation	Transportation	Transportation	Transportation	High-tech
EfCI	1.772	1.772	1.772	1.772	1.350	0.664

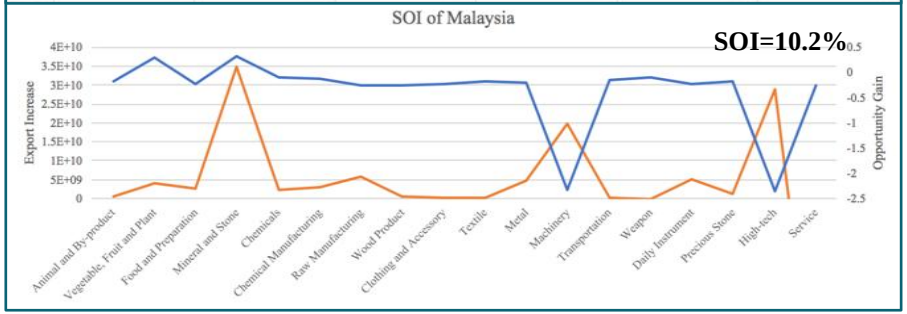


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	1.420	0.000
Incremental	0.351	-0.133

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	2.029	0.000
Incremental	0.306	-0.055

Table 6: Contingency Plan (2014)

	0	1
Machinery	Precious Stone	Weapon
High Tech	Precious Stone	Weapon

7.6 Mexico

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Vegetable, Fruit and Plant	0.290	2. Animal and By-product	0.243
3. Chemicals	**0.224**	**4. Mineral and Stone**	**0.214**
5. Metal	0.206	6. Weapon	0.199
7. Wood Product	**0.198**	**8. Clothing and Accessory**	**0.190**
9. Raw Manufacturing	0.168	10. Precious Stone	0.162
11. Chemical Manufacturing	**0.145**	**12. Transportation**	**0.140**
13. Textile	0.139	14. Food and Preparation	0.129
15. Service	**0.126**	**16. High-tech**	**0.118**
17. Machinery	0.105	18. Daily Instrument	0.045

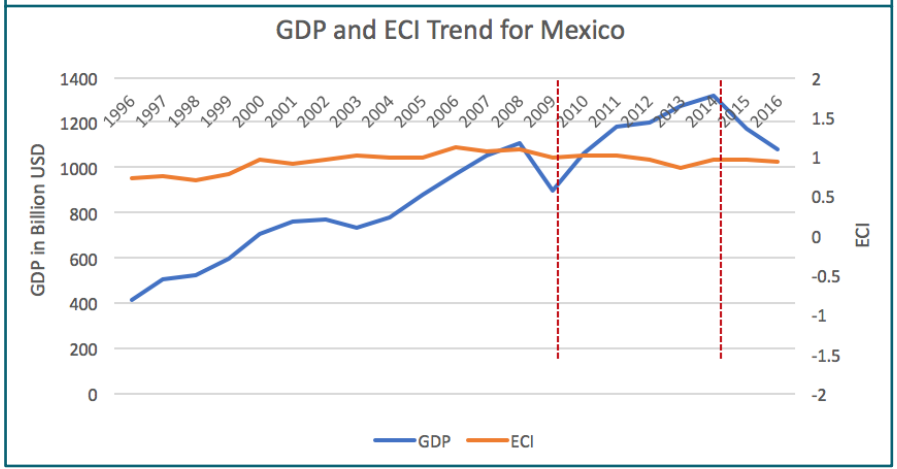


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Raw Manufacturing	-0.276
Metal	-0.197
Textile	-0.180
Wood Product	-0.178
Service	-0.122
Chemicals	-0.121
Weapon	-0.118
Chemical Manufacturing	-0.108
Clothing and Accessory	-0.090
Animal and By-product	-0.007
Vegetable, Fruit and Plant	0
Food and Preparation	0
Mineral and Stone	0
Machinery	0
Transportation	0
Daily Instrument	0
Precious Stone	0
High-tech	0

7.6 Mexico

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Vegetable, Fruit and Plant	High-tech	High-tech	High-tech	High-tech	High-tech
2	Daily Instrument	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Daily Instrument
3	Transportation	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Vegetable, Fruit and Plant
4	Machinery	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	Transportation
5	High-tech	Transportation	Transportation	Transportation	Transportation	Machinery
6	Weapon	Machinery	Machinery	Machinery	Machinery	Food and Preparation
7	Clothing and Accessory	Service	Service	Service	Service	Mineral and Stone
8	Chemical Manufacturing	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Precious Stone
9	Chemicals	Weapon	Weapon	Weapon	Weapon	Service
10	Textile	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Animal and By-product
11	Service	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Chemicals
12	Precious Stone	Chemicals	Chemicals	Chemicals	Chemicals	Clothing and Accessory
13	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Weapon
14	Mineral and Stone	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Wood Product
15	Food and Preparation	Textile	Textile	Textile	Textile	Metal
16	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product	Textile
17	Raw Manufacturing	Metal	Metal	Metal	Metal	Chemical Manufacturing
18	Metal	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
EtCI	2.901	2.344	2.344	2.357	2.357	1.756

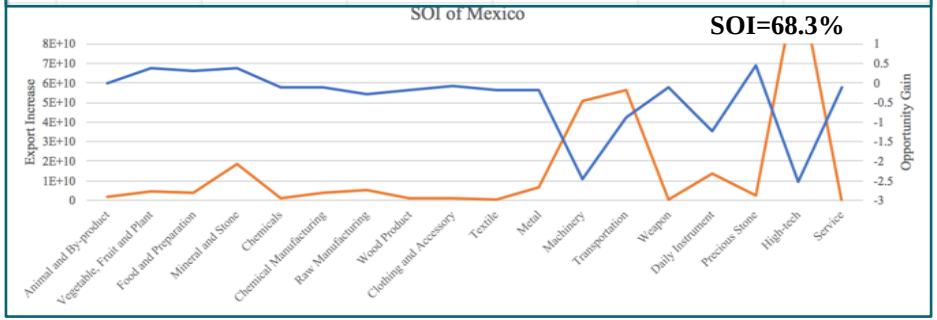


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	1.034	-0.510
Incremental	0.890	0.003

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	2.913	-0.042
Incremental	0.351	-0.093

Table 6: Contingency Plan (2009)

	0	1
High Tech	Animal and By-product	Clothing and Accessory

Table 7: Contingency Plan (2014)

	0	1
High Tech	Chemicals	Weapon

7.7 USA

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Service	0.242	2. Chemical Manufacturing	0.229
3. Food and Preparation	**0.197**	**4. Textile**	**0.196**
5. Clothing and Accessory	0.195	6. Animal and By-product	0.192
7. Metal	**0.192**	**8. Transportation**	**0.191**
9. High-tech	0.190	10. Machinery	0.184
11. Weapon	**0.183**	**12. Raw Manufacturing**	**0.181**
13. Mineral and Stone	0.171	14. Precious Stone	0.161
15. Chemicals	**0.144**	**16. Wood Product**	**0.118**
17. Daily Instrument	0.105	18. Vegetable, Fruit and Plant	0.086

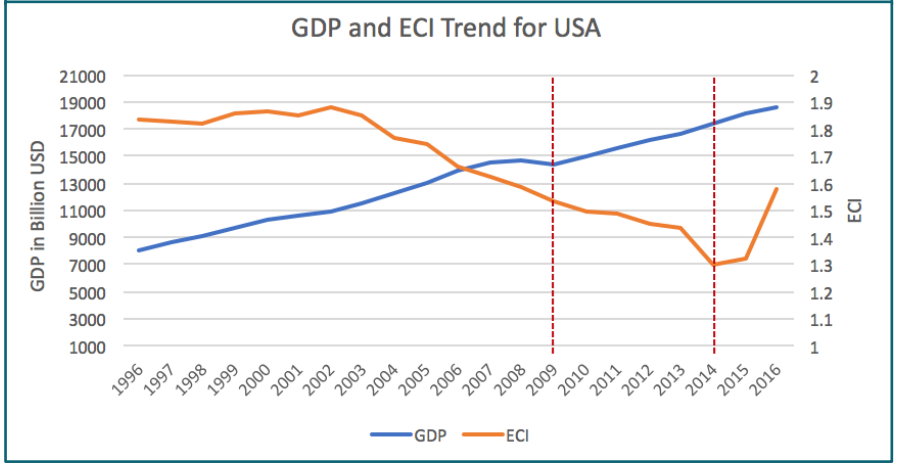


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Mineral and Stone	-0.081
Food and Preparation	-0.034
Animal and By-product	-0.021
Precious Stone	-0.002
Vegetable, Fruit and Plant	0
Chemical Manufacturing	0
Wood Product	0
Weapon	0
Daily Instrument	0
Service	0
Raw Manufacturing	0.012
Metal	0.021
Clothing and Accessory	0.044
Textile	0.096
Chemicals	0.102
Transportation	0.272
High-tech	0.382
Machinery	0.434

7.7 USA

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Machinery	Machinery	Machinery	Machinery	Machinery	Machinery
2	High-tech	High-tech	High-tech	High-tech	High-tech	Daily Instrument
3	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	High-tech
4	Transportation	Transportation	Transportation	Textile	Textile	Textile
5	Textile	Textile	Textile	Service	Service	Service
6	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Weapon	Weapon	Weapon
7	Service	Service	Service	Transportation	Transportation	Transportation
8	Weapon	Weapon	Weapon	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing
9	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Chemicals	Chemicals	Chemicals
10	Chemicals	Chemicals	Chemicals	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing
11	Metal	Metal	Metal	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory
12	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Metal	Metal	Metal
13	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product	Wood Product
14	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product	Animal and By-product
15	Food and Preparation	Food and Preparation	Food and Preparation	Precious Stone	Precious Stone	Precious Stone
16	Precious Stone	Precious Stone	Precious Stone	Food and Preparation	Food and Preparation	Food and Preparation
17	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
18	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone
EtCI	0.285	0.285	0.318	0.894	0.894	0.912

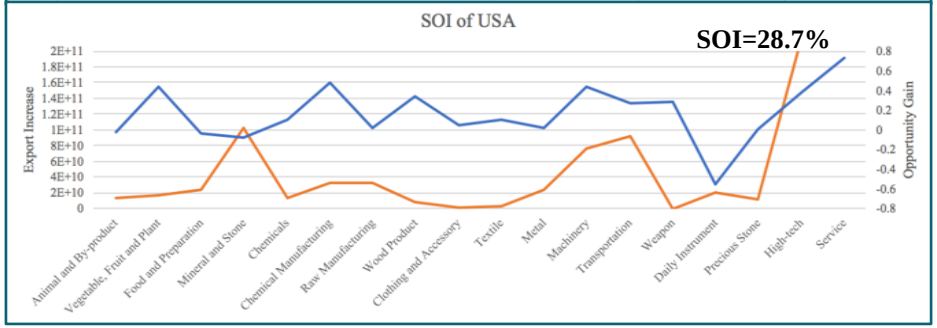


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	-0.389	-0.028
Incremental	1.104	-0.025

Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	-0.897	-0.041
Incremental	0.758	-0.113

Table 6: Contingency Plan (2009)

	0	1
Precious Stone	Machinery	High-tech
High Tech	Clothing and Accessory	Animal and By-product
Service	Clothing and Accessory	Service

Table 7: Contingency Table (2014)

	1	2
Precious Stone	Machinery	High-tech
High-tech	Machinery	Daily Instrument
Service	Mineral and Stone	Precious Stone

7.8 South Korea

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Chemicals	0.030	2. Machinery	0.028
3. High-tech	0.025	4. Weapon	0.023
5. Precious Stone	0.019	6. Mineral and Stone	0.012
7. Transportation	0.002	8. Service	-0.001
9. Raw Manufacturing	-0.002	10. Vegetable, Fruit and Plant	-0.011
11. Food and Preparation	-0.013	12. Metal	-0.022
13. Clothing and Accessory	-0.023	14. Textile	-0.032
15. Wood Product	-0.040	16. Daily Instrument	-0.048
17. Animal and By-product	-0.052	18. Chemical Manufacturing	-0.056

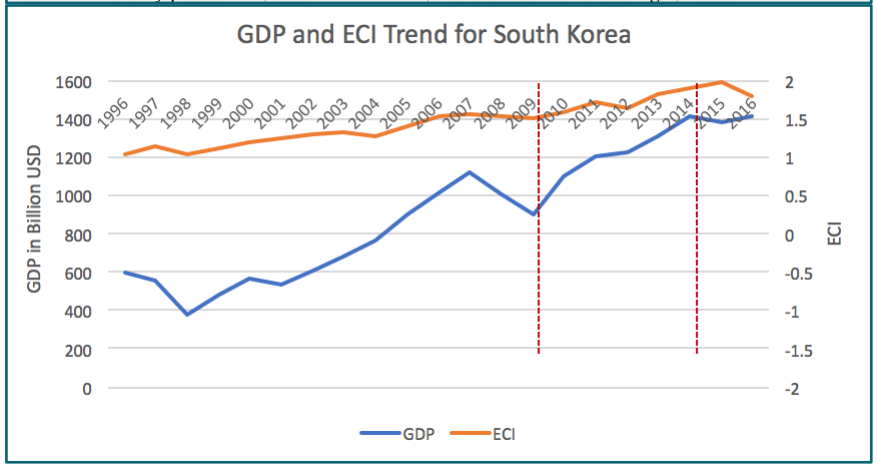


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Food and Preparation	-0.355
Vegetable, Fruit and Plant	-0.351
Wood Product	-0.337
Animal and By-product	-0.304
Precious Stone	-0.258
Service	-0.241
Mineral and Stone	-0.224
Clothing and Accessory	-0.209
Weapon	-0.170
Chemical Manufacturing	-0.160
Chemicals	0
Raw Manufacturing	0
Textile	0
Metal	0
Machinery	0
Transportation	0
Daily Instrument	0
High-tech	0

7.8 South Korea

Table 3: High-tech Sensitivity Plan Table (2014)

Ranking	0	"-10%"	"-20%"	"-30%"	"-40%"	"-50%"
1	Chemicals	Chemicals	Chemicals	Chemicals	Daily Instrument	Mineral and Stone
2	Daily Instrument	Daily Instrument	Daily Instrument	Daily Instrument	Transportation	Service
3	Transportation	Transportation	Transportation	Transportation	Machinery	Raw Manufacturing
4	Machinery	Machinery	Machinery	Machinery	Metal	Daily Instrument
5	Metal	Metal	Metal	Metal	Textile	Transportation
6	Textile	Textile	Textile	Textile	Chemicals	Machinery
7	High-tech	High-tech	High-tech	High-tech	Raw Manufacturing	Metal
8	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Raw Manufacturing	Weapon	Textile
9	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Mineral and Stone	Chemicals
10	Weapon	Weapon	Weapon	Weapon	Chemical Manufacturing	Weapon
11	Precious Stone	Precious Stone	Precious Stone	Precious Stone	Clothing and Accessory	Precious Stone
12	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Clothing and Accessory	Precious Stone	Clothing and Accessory
13	Service	Service	Service	Service	Service	Chemical Manufacturing
14	Chemical Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Animal and By-product	Animal and By-product	Animal and By-product
15	Animal and By-product	Animal and By-product	Animal and By-product	Chemical Manufacturing	Wood Product	Vegetable, Fruit and Plant
16	Wood Product	Wood Product	Wood Product	Wood Product	Food and Preparation	Wood Product
17	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation
18	Food and Preparation	Food and Preparation	Food and Preparation	Food and Preparation	High-tech	High-tech
EtCI	2.117	2.117	2.129	2.135	1.752	1.472

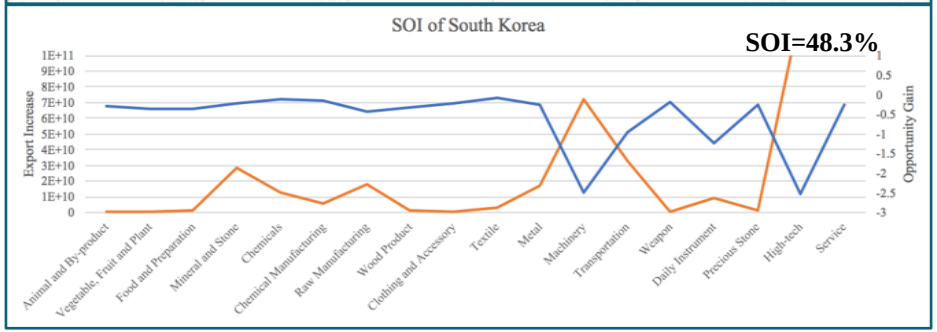


Table 4: Sensitivity Score (2009)

	High Tech	Food and Preparation
Decremental	0.901	0.000
Incremental	0.326	0.000

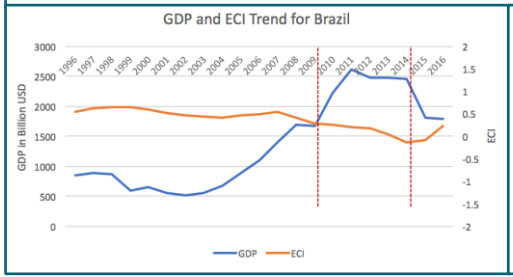
Table 5: Sensitivity Score (2014)

	High Tech	Food and Preparation
Decremental	1.202	0.000
Incremental	1.081	-0.002

7.9 Brazil

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Daily Instrument	0.157	2. High-tech	0.156
3. Machinery	**0.125**	**4. Metal**	**0.094**
5. Chemical Manufacturing	0.088	6. Weapon	0.087
7. Textile	**0.083**	**8. Service**	**0.065**
9. Wood Product	0.063	10. Transportation	0.059
11. Animal and By-product	**0.057**	**12. Mineral and Stone**	**0.056**
13. Food and Preparation	0.046	14. Raw Manufacturing	0.018
15. Chemicals	**0.009**	**16. Vegetable, Fruit and Plant**	**0.004**
17. Precious Stone	0.003	18. Clothing and Accessory	0.001



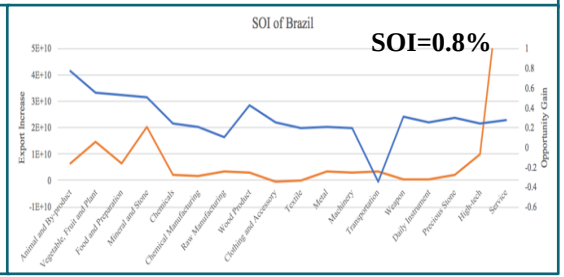


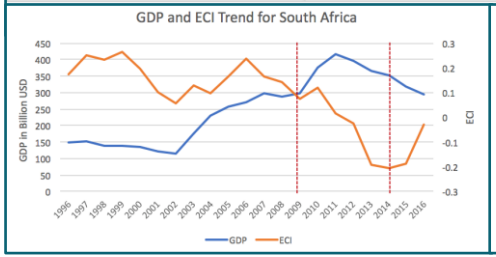
Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Animal and By-product	0
Vegetable, Fruit and Plant	0
Food and Preparation	0
Mineral and Stone	0
Chemicals	0
Raw Manufacturing	0
Wood Product	0
Metal	0
Transportation	0
Weapon	0
Textile	0.194
Machinery	0.198
Chemical Manufacturing	0.211
High-tech	0.237
Daily Instrument	0.253
Clothing and Accessory	0.256
Service	0.276
Precious Stone	0.300

7.10 South Africa

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Animal and By-product	0.056	2. Raw Manufacturing	0.045
3. Daily Instrument	**0.037**	**4. Food and Preparation**	**0.033**
5. Vegetable, Fruit and Plant	0.030	6. Textile	0.022
7. High-tech	**0.021**	**8. Clothing and Accessory**	**0.016**
9. Chemical Manufacturing	0.010	10. Transportation	0.005
11. Wood Product	**0.002**	**12. Metal**	**-0.001**
13. Weapon	-0.003	14. Machinery	-0.004
15. Mineral and Stone	**-0.012**	**16. Service**	**-0.020**
17. Chemicals	-0.038	18. Precious Stone	-0.059



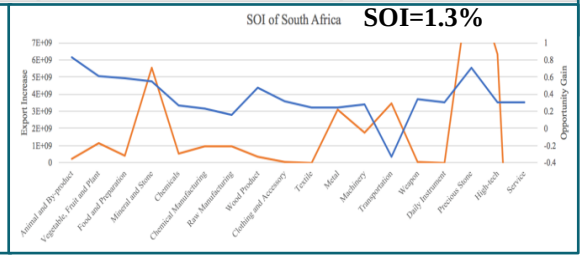


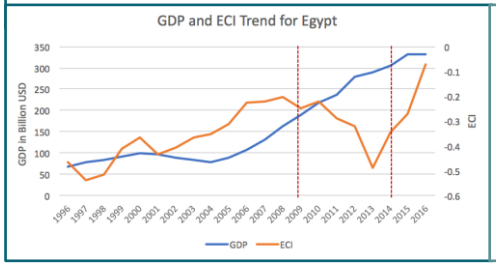
Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Animal and By-product	0
Vegetable, Fruit and Plant	0
Food and Preparation	0
Mineral and Stone	0
Chemicals	0
Wood Product	0
Metal	0
Transportation	0
Weapon	0
Precious Stone	0
Raw Manufacturing	0.152
Chemical Manufacturing	0.232
Textile	0.238
Machinery	0.282
Daily Instrument	0.299
High-tech	0.300
Service	0.300
Clothing and Accessory	0.315

7.11 Egypt

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Animal and By-product	0.324	2. Service	0.298
3. Weapon	**0.296**	**4. Vegetable, Fruit and Plant**	**0.294**
5. Mineral and Stone	0.292	6. Wood Product	0.291
7. Clothing and Accessory	**0.291**	**8. Textile**	**0.270**
9. Raw Manufacturing	0.261	10. Chemicals	0.247
11. Daily Instrument	**0.239**	**12. Precious Stone**	**0.224**
13. Chemical Manufacturing	0.211	14. Transportation	0.210
15. Food and Preparation	**0.201**	**16. Metal**	**0.200**
17. High-tech	0.135	18. Machinery	0.132



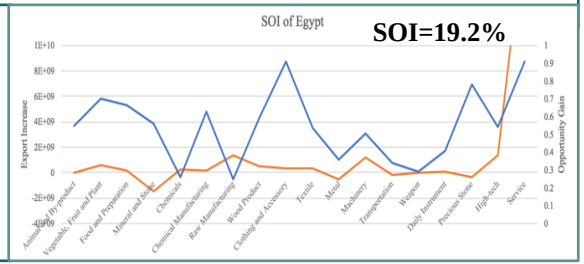


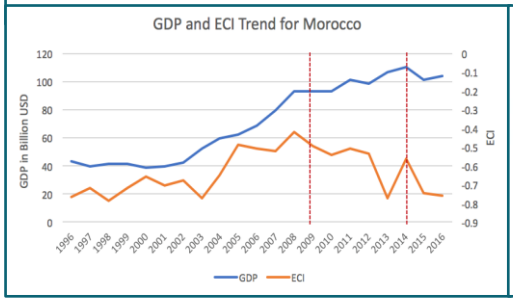
Table 2: Suggested Development Vector (SDV)

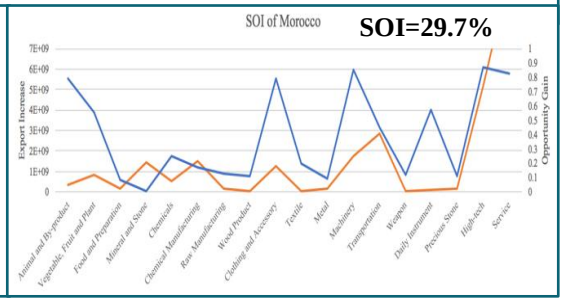
Product	SDV Value
Vegetable, Fruit and Plant	0
Food and Preparation	0
Mineral and Stone	0
Chemical Manufacturing	0
Wood Product	0
Clothing and Accessory	0
Textile	0
Metal	0
Precious Stone	0
Service	0
Raw Manufacturing	0.246
Chemicals	0.259
Weapon	0.292
Transportation	0.337
Daily Instrument	0.409
Machinery	0.508
High-tech	0.543
Animal and By-product	0.545

7.12 Morocco

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Chemical Manufacturing	0.024	2. Precious Stone	-0.029
3. Wood Product	**-0.031**	**4. Textile**	**-0.046**
5. Clothing and Accessory	-0.050	6. Vegetable, Fruit and Plant	-0.057
7. Metal	**-0.058**	**8. Food and Preparation**	**-0.059**
9. Animal and By-product	-0.059	10. Chemicals	-0.061
11. Service	**-0.063**	**12. Mineral and Stone**	**-0.065**
13. Raw Manufacturing	-0.069	14. High-tech	-0.069
15. Daily Instrument	**-0.073**	**16. Transportation**	**-0.073**
17. Machinery	-0.076	18. Weapon	-0.087





SOI=29.7%

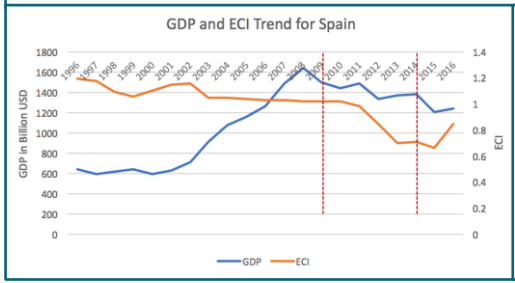
Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Animal and By-product	0
Vegetable, Fruit and Plant	0
Chemicals	0
Clothing and Accessory	0
Service	0
Mineral and Stone	0.007
Food and Preparation	0.087
Metal	0.093
Precious Stone	0.110
Wood Product	0.111
Weapon	0.116
Raw Manufacturing	0.129
Chemical Manufacturing	0.167
Textile	0.201
Transportation	0.448
Daily Instrument	0.568
Machinery	0.848
High-tech	0.871

7.13 Spain

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Daily Instrument	-0.261	2. High-tech	-0.278
3. Weapon	**-0.288**	**4. Chemical Manufacturing**	**-0.288**
5. Machinery	-0.309	6. Wood Product	-0.314
7. Food and Preparation	**-0.317**	**8. Transportation**	**-0.323**
9. Chemicals	-0.332	10. Metal	-0.344
11. Textile	**-0.351**	**12. Raw Manufacturing**	**-0.356**
13. Clothing and Accessory	-0.388	14. Precious Stone	-0.391
15. Animal and By-product	**-0.393**	**16. Service**	**-0.404**
17. Vegetable, Fruit and Plant	-0.418	18. Mineral and Stone	-0.431



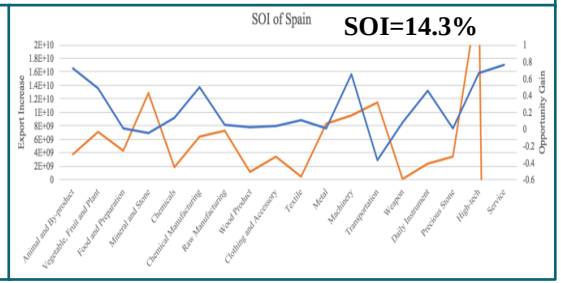


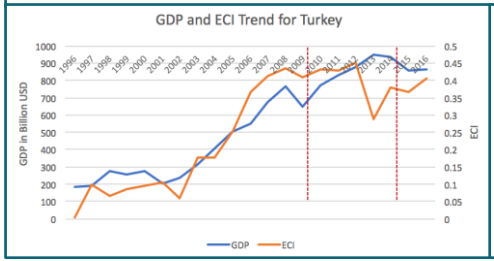
Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Mineral and Stone	-0.052
Animal and By-product	0
Vegetable, Fruit and Plant	0
Chemical Manufacturing	0
Transportation	0
Service	0
Precious Stone	0.004
Metal	0.008
Food and Preparation	0.009
Wood Product	0.029
Clothing and Accessory	0.042
Raw Manufacturing	0.051
Weapon	0.072
Textile	0.109
Chemicals	0.131
Daily Instrument	0.458
Machinery	0.649
High-tech	0.659

7.14 Turkey

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Daily Instrument	-0.093	2. Machinery	-0.153
3. Chemicals	**-0.153**	**4. High-tech**	**-0.162**
5. Chemical Manufacturing	-0.177	6. Wood Product	-0.180
7. Food and Preparation	**-0.184**	**8. Raw Manufacturing**	**-0.191**
9. Textile	-0.192	10. Weapon	-0.201
11. Transportation	**-0.219**	**12. Metal**	**-0.229**
13. Clothing and Accessory	-0.232	14. Precious Stone	-0.245
15. Service	**-0.254**	**16. Animal and By-product**	**-0.272**
17. Mineral and Stone	-0.281	18. Vegetable, Fruit and Plant	-0.303



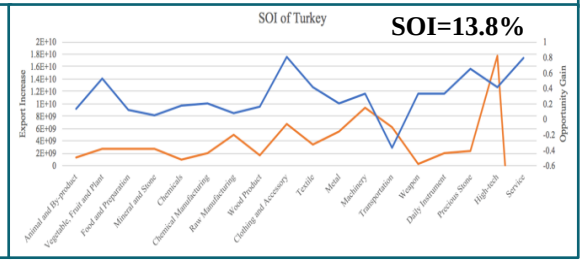


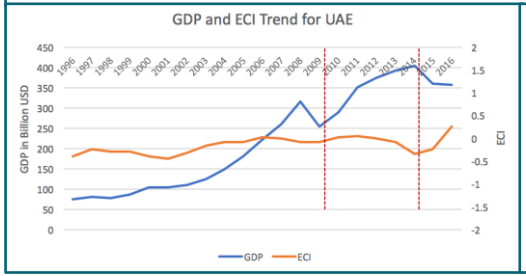
Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Vegetable, Fruit and Plant	0
Clothing and Accessory	0
Textile	0
Metal	0
Transportation	0
Weapon	0
Precious Stone	0
Service	0
Mineral and Stone	0.054
Raw Manufacturing	0.073
Food and Preparation	0.118
Animal and By-product	0.139
Wood Product	0.160
Chemicals	0.176
Chemical Manufacturing	0.202
Daily Instrument	0.335
Machinery	0.335
High-tech	0.407

7.15 United Arab Emirates

Table 1: Actual Development Index (ADI)

Category	ADI	Category	ADI
1. Raw Manufacturing	0.057	2. Animal and By-product	0.053
3. Food and Preparation	**0.042**	**4. Textile**	**0.041**
5. Clothing and Accessory	0.037	6. Daily Instrument	0.030
7. Vegetable, Fruit and Plant	**0.029**	**8. Metal**	**0.028**
9. High-tech	0.028	10. Machinery	0.027
11. Service	**0.025**	**12. Wood Product**	**0.012**
13. Weapon	0.011	14. Precious Stone	0.002
15. Mineral and Stone	**-0.002**	**16. Transportation**	**-0.009**
17. Chemical Manufacturing	-0.011	18. Chemicals	-0.020



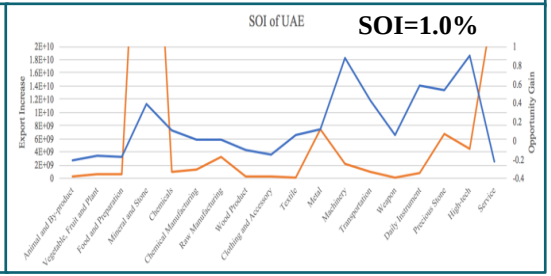


Table 2: Suggested Development Vector (SDV)

Product	SDV Value
Service	-0.226
Animal and By-product	-0.210
Food and Preparation	-0.177
Vegetable, Fruit and Plant	-0.161
Clothing and Accessory	-0.144
Wood Product	-0.102
Mineral and Stone	0
Metal	0
Precious Stone	0
Raw Manufacturing	0.004
Chemical Manufacturing	0.010
Textile	0.053
Weapon	0.058
Chemicals	0.114
Transportation	0.426
Daily Instrument	0.581
Machinery	0.883
High-tech	0.908